

PROPOSAL PART I: KEY INFORMATION

- All of the items in Part I “Key Information” section will be inserted as the opening pages of the Phase I proposal.
- Do not number the pages in Part I. (Numbering should begin on the first page of the “Project Narrative-Technical Content.”)
- To structure Part I of your proposal, follow the exact format as the items below.
- No other cover pages are permitted to precede Part I.
- Proposals that do not include a Part I section exactly as specified below will be rejected without review.

Part I: Key Information

Section A.

- A) Project Title: Updating HELP Math using HTML5
- B) Name of the small business: Boulder Learning Inc.
- C) DUNS: 040574427
- D) Small Business Address and Phone: 269 Sam Grange Ct, Carbondale, CO 81623-7716; 970 704-1465
- E) Company Website URL: <https://boulderlearning.com/>
- F) For the Principal Investigator, fill in the below table:

Name	Address	Phone Number	Email
Brandon Holding	2900 S 46th St, Lincoln, NE 68506	480 829 9678	bholding@boulderlearning.com



X

Authorized Signature

- G) For the representative authorized to represent the small business concern in negotiations, fill in the below table:

H) Name	Address	Phone Number	Email
John Ramo	269 Sam Grange Ct. Carbondale, CO 81623	970-948-7062	jramo@boulderlearning.com



X

Authorized Signature

- l) For all key members of the project team in the table below (e.g., Principal Investigator, Researcher, Developer, Administrative Assistant), fill in the below table. *(Add lines as needed)*

Name	Professional Affiliation (list only one affiliation, whether at the small business or another organization, or as an independent consultant)	Project Role	Email
Brandon Holding	Boulder Learning, Inc.	PI	bholding@boulderlearning.com
John Ramo	Boulder Learning, Inc.	co-PI	Jramo@boulderlearning.com
Linda Hardman	Independent Consultant	Learning Architect	jhardman@austin.rr.com
Megan Schmeister	Boulder Learning, Inc.	Product Manager	megan@helpprogram.net
Venkatesh Venkataramani	Independent Consultant	Program Analysis	mvssofttech@gmail.com

(Add additional lines for more team members)

- J) Indicate whether the proposal is for the R&D of an entirely new product, for R&D to add new and unique components to a prototype that already exists and is functioning, or for R&D to add new and unique components to a product that is already fully developed.

(Select only one.)

☐ for R&D of an entirely new product where no (or very limited) previous technological development work has occurred

☐ for R&D to add new and unique components to a prototype of an education technology product that already exists and is functioning

X for R&D to add new and unique components to an educational technology product that is already fully developed

K) Video Demonstration: For proposals to develop a new component to an existing prototype or a fully developed product that already exists and is functioning, the offeror may include a URL (Uniform Resource Locator) link to a video demonstration that presents the current version to a reviewer.

Specifications:

- *Video Demonstrations are optional to include in the proposal. If a video demo is included, reviewers will view the video as a component of the proposal.*
- *All Video Demonstrations should be posted to YouTube (<http://www.youtube.com/>) prior to the submission deadline, and available during the review period up to 90-days after.*
- *If the Video Demonstration is posted on a website other than YouTube, it is possible reviewers will not be able to watch the video. Videos can be no longer than 4 minutes in duration. Reviewers will be instructed to stop viewing at 4 minutes.*
- *Video Demonstrations are not permitted to be updated after submission. If the posting date on YouTube for the video is later than the submission date, the video will be deemed ineligible for inclusion in the review process.*
- *Offerors are not required to provide a video for proposals for an entirely new product or where no (or limited) previous technological development work has occurred, although offerors may provide a video on related prior research or development project if they choose.*

Insert a URL to a video demonstration here:	http://boulderlearning.com/products/helpmath/
--	---

L) Priority. Select one priority area. (*Note: While the focus of a proposal may be applicable to more than one priority area – offerors must select the priority area that is the best fit.*)

- **PRIORITY AREA 2: Products to Modernize Assessment and Support Student Learning and Relevant Outcomes**

See Section IX for additional information on the Priority Areas.

M) **Abstract.** Provide an abstract with the following two components. (*Nothing in this section should be proprietary or confidential.*)

- Technical abstract limited to 200 words.

Boulder Learning, Inc. has, as part of its suite of educational tools, a program entitled *HELP Math* (HP). Currently, HELP Math is limited to Java platforms using the Chrome browser. This project will extend HP to include HTML5 such that it can and will run on a variety of browsers, PCs, Apple, and mobile technologies. Historically, HP has helped students that struggle with mathematics, and done so with multimedia technologies, interactive storylines, and at the pace of students as they learn. In fact, HP was featured as a member of the What Works Clearinghouse (https://ies.ed.gov/ncee/wwc/Docs/SingleStudyReviews/wwc_help_102312.pdf)

HP, however, has been limited by the types of computers and browsers available to students that struggle in schools. The help that HELP Math offers, is therefore, similarly limited. To remedy this situation the extant HELP Math program will be updated so that it works across a wide set of platforms, programs, and is thereby that much more effective. HTML5 has been demonstrated by Google as enhancing the productivity, creativity, engagement, browser compatibility, more effective and fluent video playback, and is widely available to public commercial enterprises. Accordingly, we will use it here.

- A summary of the commercial potential of the product or tool if development were completed as described above limited to 100 words.

We reviewed the target market in the US in terms of its structure and size. States are responsible for the education of their residents. These states have 13,588 school districts. According to statistics compiled by various agencies and organizations, there are over 132,181 operational public elementary and secondary schools in the U.S. states and DC. If every school bought one HELP program for each grade level, K-8, revenue would increase nine fold, to \$675 million.. A 1% share of this market is \$6.75 million annually.

PROPOSAL PART II. Project Narrative – Technical Content

1. Significance

a. Problem

Simply put, the problem that we face as STEM education professionals and stakeholders, and perhaps as a larger social system, is understanding efficient, effective methods with which to help children learn. In this proposal, *Updating HELP Math using HTML5*, Boulder Learning, Inc. and the University of Nebraska-Lincoln are partnering to add new components and capacities to an already-developed, and demonstrably efficacious (Drew Polly & Chandra Orrill, 2012) educational software platform: HELP Math (HM). Specifically, we will address the IES “PRIORITY AREA 2: Products to Modernize Assessment and Support Student Learning and Relevant Outcomes”.

Previous versions of HM have been limited by the types of computers and browsers available to students that struggle in schools with both language and mathematics. The help that HELP Math offers, is therefore, similarly limited. To remedy this situation, the extant HELP Math program will be updated so that it operates across a wide variety of platforms, programs, and is thereby significantly more effective using HTML5. HTML5 has been linked to enhancing productivity, creativity, engagement, browser compatibility, more effective and fluent video playback, and is widely available to public commercial enterprises (Haag, 2011). This is especially important to ensure that HM can be used across mobile platforms and is that much more accessible to students, as part of a future wherein mobile technologies become more and more dominant, especially in shaping young students’ experiences (Bond, 2014; David, 2012). HTML5 as a justifiable innovation is of course not enough to warrant this proposal. The focus on middle school, rational number mathematics, and language acquisition is also pertinent.

Overall, there is a continuing clear and urgent need in the U.S. to improve mathematics instruction and academic achievement of English Language Learners (ELLs) and other struggling students in grades K-8 (Viesca et al., 2019). Previous version of HM as well as the increased capacity for it in this proposal focuses on middle school arithmetic (which includes ratios/proportional reasoning) and rational number understanding of concepts and skills. This is a critical topic for young learners for several reasons. First, it is the foundation of success in middle school mathematics (Beyranvand, 2014) and how classroom instruction addresses national standards (CCSS, 2010). Second, early arithmetic is strongly connected to and predictive of success in rational number learning (Behr et al., 1992). Third, simple arithmetic is correlated with interest in STEM careers (Baluch & Page Baluch, 2016). Indeed, HP helps students learn arithmetic and needs to be updated. This also builds on best practices associated with arithmetic and multimedia learning (this is discussed in greater detail in other sections of the proposal) (R. E. Mayer, 2005).

The combination of a focus on middle and young students’ understanding of arithmetic, using intelligent, effective online media, and the need for that media to be accessible across a variety of platforms and operating systems generates a clear and present mandate for the work described below.

b. The product, its implementation, and the intended outcomes

HM is meant to help struggling 6-8th grade students. Citing a previous efficacy study using HM as an intervention (Crawford, Schmeister, & McDonald, 2010), HM is:

a web-based supplementary math curriculum designed for ELL students by employing sheltered instruction to deliver math lessons with interactive Spanish support. HELP Math is comprised of 44 lessons containing over 100 hours of instruction targeting grades six through eight. The four modules include: Numbers Make Sense, Geometry, Algebra, and Data Analysis (pg. 15).

A brief overview of HM can be viewed at <http://boulderlearning.com/products/helpmath/>.

HM will be supplemented by re-coding using HTML5. HTML5 is the updated Hypertext Markup Language (HTML), a standard programming language for describing the contents and appearance of the web pages and its elements. This will ensure cross browser compatibility and will work with CSS3 (cascading style sheet). At present all major browsers support HTML5 tags and will render the pages correctly. Secondly, HTML5 is an improved alternative to traditional flash applications because unlike flash, it does not require an adobe player installed on the user machine. This is important for adaptations of HTML5 in mobile applications and game development. Lastly, HTML5 will include video and audio integrations, without needing to rely on third-party plugins.

The same outcomes previously obtained by using HM are expected only at greater scale with HTML5. This is because the efficacious product of HM will be that much more accessible to struggling students and their teachers.

c. Theoretical and empirical support

Broadly, HP works because it draws on principles of effective multimedia learning as cited in extant research. For example, Lemke (1998) compellingly argued that curricular demands for maximal learning required students to “integrate specialized verbal, visual, and mathematical literacies quickly and fluently in real time” (pg. 247). Moreover, a recent synthesis of research on physical and virtual laboratories in science and engineering education (and embedded mathematics within those virtual laboratories) by de Jon, Linn & Zacharia (2013) concluded that a) virtual experiments can provide equivalent and often superior learning relative to physical experiments and the mathematics within it, and b) combining physical manipulation of virtual media consistently provided superior learning relative to physical experiments alone. Lastly, several studies have shown greater learning gains when students were able to view or interact with simulations of unobservable phenomenon (Olympiou et al., 2013). We draw on these principles in HP and how students interact with the mathematics content. This harkens to a larger learning principle: the modality principle.

Modality Principle: The design of one-on-one media tutoring in science and math is based on a strong foundation of learning theory and evidence in multimedia learning (R. Mayer, 2005; R. E. Mayer, 2001, 2005). The key finding that motivates the use of multimedia interactive experiences in learning (with narrated animations) is the modality principle. This is derived from research indicating that simultaneously presenting visual information with spoken narration/animations results in the highest retention of information and transfer of knowledge to new tasks (ibid). More specifically, according to the Cognitive Theory of Multimedia Learning, a student’s limited capacity for auditory and visual inputs are processed independently, and then integrated to produce enriched mental representations. HP draws on these learning principles, and model of cognition in design that has led to its multiple successes and acceptance into the What Works Clearinghouse.

Other principles include segmentation, pacing, and relatively short intervals that focus on single concepts (Wouters et al., 2005). This ensures that sensory input is not overloaded and that learning can be optimized without saturation (Brünken et al., 2002; Sweller & Chandler, 1991). This allows HM to build its successful activities and learning outcomes such that it corresponds with best practices in multimedia learning in curriculum and instruction as well as psychology learning theories. Specifically, it draws on sheltered scaffolding.

Sheltered Instruction and Scaffolding and Techniques Underpinning HELP Features are described below in Table 1.

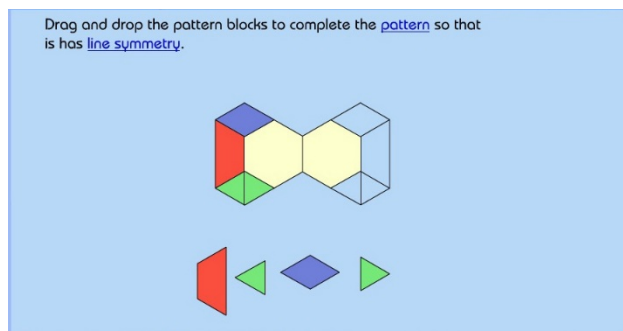
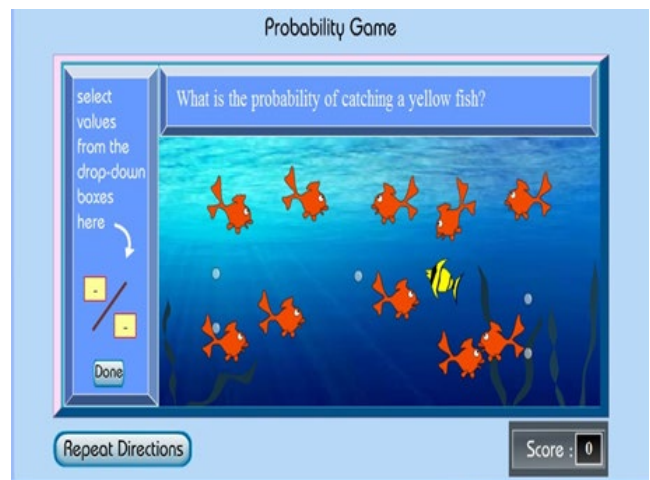
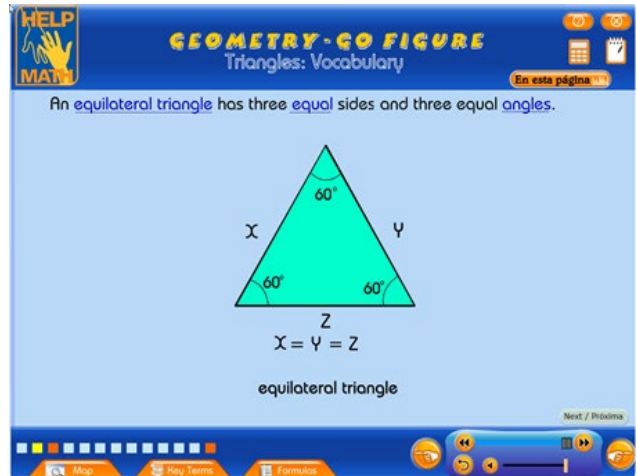
Table 1: Sheltered Instruction & Scaffolding Techniques Underpinning HELP Features

Key Sheltered Instructional Principle	HELP Feature
Increase content comprehensibility using extra-linguistic cues (Krashen, 1992)	Synchronizes audio, visual, text, and interactivity (every page) to create a visual connection between words and meaning. For example, corresponding vocabulary, symbols or pictures flash in sync with audio.
Direct, targeted vocabulary development: technical & academic English (Collier, 1987; Cummins, 1979, 1981; Marzano, 2004)	Explicitly teaches vocabulary includes a bilingual & pictorial dictionary & contextual hyperlinks (every page). Includes key concept vocabulary (fractions, tessellation), supporting vocabulary (numerator, plane) & academic English (simplify the expression, which of the following, if not X then Y).
Build background concept knowledge and pre-requisite skills (Marzano, 2004)	Provides comprehensive foundational lessons and tools to individualize & customize lessons and curriculum. Both grade level and prior grade content is always easily accessible.
Review and assess throughout with frequent comprehension checks. (Echevarria & Short, 1999; Heward, 2009)	Provides immediate corrective and encouraging, positive feedback, consistent throughout all lessons and quizzes. HELP interactivity with unobtrusive testing and feedback is provided shortly after each instruction and demonstration.
Break down math procedures and concepts into small “learning chunks”	Demonstrates procedural steps and algorithms associated with problem solving.
Home language support and translations to make content comprehensible (Halle et al., 2011)	Provides Spanish audio and full bilingual translation (every page).
Recycle prior knowledge & connect what students already know with what they are learning (Echevarria & Short, 1999)	Provides visual & contextual hints and clues throughout the program and “Need More HELP” buttons on every “Try-It” page. HELP never allows a student to “get stuck”.

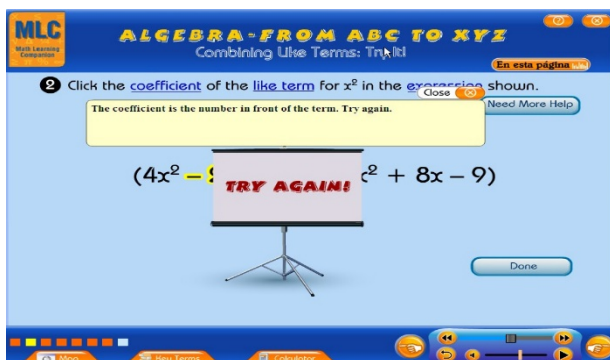
Increase connections to student lives (Echevarria & Short, 1999)	Provides “Real World” scenarios at the start of each lesson.
Increase Higher Order Thinking Skills (Echevarria & Short, 1999)	Explicitly teaches problem solving approaches & test-taking skills – how to read, comprehend and answer ‘look alike’ questions from state standardized math tests.
Repetition and reinforcement of concepts, procedures and vocabulary through interaction (Echevarria & Short, 1999)	Incorporates games, puzzles and interactivities throughout all lessons (e.g. number lines, base10 blocks, drag & drop, money, matching, etc.) which consistently repeat, vary and contextualize the math concepts and vocabulary.
Repetition and reinforcement (Echevarria & Short, 1999)	Provides manipulatives available throughout the program (e.g. calculator, protractors, rulers, fraction strips & wheels, thermometers, etc.) with structured and patterned activities around clear, embedded lesson objectives.
Scaffolds the learner by providing a cognitive support to move student just beyond what they can learn unassisted (Vygotsky & Cambridge, 1978; Wood et al., 1976)	In addition to all of the above techniques, HELP is also replete with music, animation, color, providing a non-threatening environment for practice, knowledge construction, and “play”. The “math coach” in the audio speaks at a moderate rate with mild and encouraging tone. If the student does not understand, they can replay the audio or look up the Key Terms which they do not understand or want a visual example of the term.
Scaffolds the mathematics content instruction (ibid) and simultaneous language acquisition (Echevarria & Short, 1999)	Clear math and language objectives for each lesson, embeds NCTM focal points into lesson objectives & course structure, sequencing problems/concepts to progress from easy to more difficult, and procedural and problem solving strategies develop from teacher (program) modeled to student-directed.

d. Related R/R&D by the project team

The What Works Clearinghouse gave HELP Math its highest possible rating (*WWC Review of the Report “Help with English Language Proficiency ‘HELP’ Program Evaluation of Sheltered Instruction Multimedia Lessons,”* n.d.). It was based on a study citing statistically significant higher scores with Digital Direction’s research-based intervention (Score, n.d.). HM is a web-based intervention designed to develop math proficiency and increase academic language mastery. Actual screenshots are provided below in Figures 1-11 below.



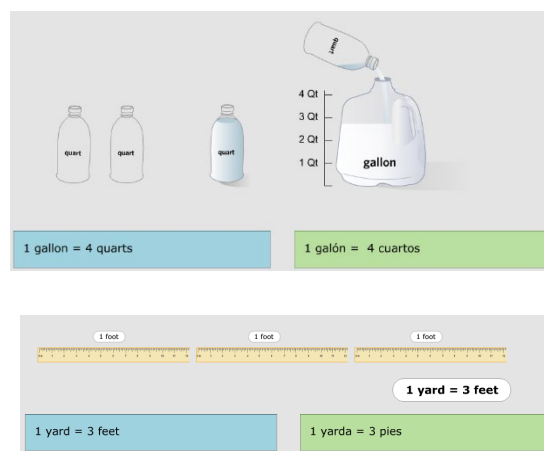
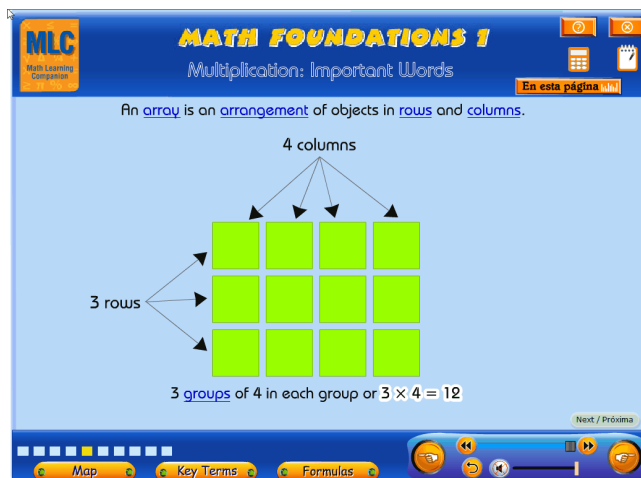
- | | |
|------------------------------|-----------------------|
| 1. Restate the question. | 3. Solve the problem. |
| 2. Organize the information. | 4. Check your work. |



Sara bought $\frac{7}{8}$ yard of fabric to make a pillow. She only used $\frac{2}{3}$ of that fabric. How much fabric did Sara use to make the pillow?

- How much fabric did Sara use?
- Information given: Sara bought $\frac{7}{8}$ yard; she used $\frac{2}{3}$ of it. Information needed: how much fabric she used
- Use strategy: Solve the equation $\frac{7}{8} \times \frac{2}{3} = ?$.
Cancel any common factors.
 $\frac{7}{\cancel{4}} \times \frac{\cancel{2}}{3} = ?$
Multiply the numerators.
 $7 \times 1 = 7$
Multiply the denominators.
 $\cancel{4} \times 3 = 12$
 $\frac{7}{12} \times \frac{2}{3} = \frac{7}{12}$ yard
- Use strategy: Draw a Picture.
 $\frac{7}{8} \times \frac{2}{3} = \frac{14}{12} = \frac{7}{6}$
 $\frac{14}{12}$ or $\frac{7}{6}$ yard was used.

A $\frac{7}{12}$ yard
 B $\frac{9}{12}$ yard
 C $\frac{9}{11}$ yard
 D $\frac{8}{7}$ yard



It is aligned with the Common Core State Standards (Drew Polly & Chandra Orrill, 2012), making it excellent preparation for the content knowledge assessed by PARCC (Partnership for Assessment of Readiness for College and Careers) (*Partnership for Assessment of Readiness for College and Careers - SAGE Research Methods*, n.d.), SBAC (Smarter Balanced Assessment Consortium) (Smarter Balanced Assessment Consortium, 2010), etc. HM can be viewed at <https://boulderlearning.com/products/helpmath/>. HM nevertheless requires updating using HTML5 for several reasons.

First, HM was originally created using Adobe Flash, which has become a legacy application and is no longer supported by many, major web browsers. This limits the usefulness of HM. Second, most modern browsers are explicitly stopping flash content, especially on mobile platforms, which requires new forms which are more compatible. In this case, that new form is HTML5. HTML5 has as its greatest advantage developing HM once and be able to deliver it across platforms and browsers. It can then be delivered in the same way to all users and their computers: desktops, tablets and smartphones, etc. This will increase the number of users of HM as well as enhance its effectiveness. The earlier efficacy study included compatible browsers and computers. When those were available, HM was demonstrably efficacious. For that efficacy to be realized as effectiveness it needs to be scalable across platforms, browsers, computers, and thereby users writ large.

e. Uniqueness of proposed product & description of similar products or typical practices and associated costs

The following pedagogical features make HM unique because they allow the program to be used by all students trying to reach proficiency in Math while acquiring both academic and mathematical language. HM has been used successfully by struggling students with Dyscalculia and students on the Autism Spectrum as well as ELLs who find the Spanish supports and the emphasis on Academic Vocabulary especially unique.

Table 2: Unique Features of HM

Unique Aspect	Unique Features
Focus on ensuring mastery of foundation skills	HELP Math delivers comprehensive instruction in all foundation skills of math; whole numbers, fractions and decimals, concepts and skills, number sense, operations, as well as rational numbers for higher grades. Emphasis is placed on key math milestones: counting, adding, subtracting, grouping, place value, properties of operations, measuring (lengths, time, money), shapes and patterns, and magnitude.
Explicit and systematic instruction	HELP Math provides explicit and sequential instruction throughout the lessons, ensuring identification and remediation of any skills deficits. Concepts are broken down into small learning “chunks,” enabling increased comprehension.
Provide models of problem solving, guided practice, constant feedback, frequent review	A variety of scaffolds and descriptive feedback loops are available in HELP Math, along with built-in review checks. Concept and skill development progresses from easy to more difficult.
Use visual representations	As HELP Math is built around sheltered instruction strategies, all math concepts are represented visually and auditorily.
Fluent retrieval of facts (rec. 10 min/day)	Fluency practice is built into all HELP Math lessons.
Frequent progress monitoring	HELP Math provides teachers with diagnostic tests to determine students’ and class mastery and skills deficits, as well as pro-actively notifying teachers when students continue to struggle and recommending further targeted interventions.
Include motivational strategies in tier 2 and tier 3 interventions	HELP Math utilizes a variety of engaging strategies to deliver instruction and ensure mastery, using a range of virtual manipulatives and tools, such as timelines, fraction wheels, place value charts, hundreds charts, protractor, calculator, thermometer, and includes learning games in each lesson.

f. Potential commercial application

Our commercialization plan is based on a vision of the near future in which we are able to offer a redeveloped HELP Math program as an updated (HTML5), supplemental, comprehensive product for students to learn K-8 math and reading, especially struggling language and math students. It will leverage over a decade of usage by students in 28 states and a research base that was accepted “without reservation” by the What Works Clearinghouse. Also, upon full funding for redevelopment, HELP Math will provide immersive experiences through technology innovations that represent a new generation of intelligent tutoring systems. We will provide teachers and students with unique, efficient and effective grades K-8 learning tools, with the primary goal of improving students’ agency and achievement in math and reading.

Our commercialization effort is based on a considered plan. The product, HELP Math will be enhanced by an upgrade to HTML5 to make the program accessible to all browsers and platforms and new innovations which will be proposed for the SBIR Phase II project (HELP Math 2.0+R). It will be marketed as a supplemental, comprehensive solution to improve elementary students' math and reading achievement. This program will be unique in the integration of math and reading, aligned with the mission of the Framework Council of Great City Schools (Council of the Great City Schools, 2014) and will use Boulder Learning's proprietary speech recognition technologies with a goal to integrate conversational interaction with emotionally intelligent virtual tutors using Machine Learning and Deep Learning.

Initially, HM was funded in 2005 under a USDE Ready to Teach Grant and since then, has won many awards and has been successfully marketed and sold to schools in 28 states. We endeavor to re-enter the K-8 math market with Help Math 2.0+R designed to improve math learning and academic vocabulary and reading fluency after first ensuring it is usable by all students through the applicability of HTML5. In sum, there is an urgent and critical need to improve reading and math achievement, in US schools and our innovations have the potential to provide teachers with a highly efficient, adaptive and affordable tool for improving students' agency and learning.

Industry Overview and Partners

Over the past two decades all major education publishers have undertaken a major shift towards digital publishing. For example, Pearson, the world's largest education publisher, reports that a major portion of its revenue is based on sales of digital content. Today, the vast majority of digital learning systems use the Software as a Service (SaaS) model, in which students interact with learning programs running on cloud-based servers using computers or mobile devices within their schools. According to Resnick (2005), about one-half of all digital content delivered to K-12 classrooms in US schools is provided by "small" companies, with annual revenues under \$10 million, with the vast majority of these companies having revenues under \$2 million. Our commercialization strategy is to market HELP Math to schools throughout the US through an expansion of Boulder Learning's current strategic partnership with Dallas based Istation, a leading provider of digital education resources to over 5 million students annually. Additionally, an obvious strategy for generating initial revenue for our innovation is to market and sell HM to schools that have previously used HELP Math with only Flash but need an HTML5 version to renew subscriptions. As mentioned previously, Istation currently serves over 5 million students in schools across the U.S. CEO, Richard Collins, has agreed to participate in our efforts.

About Istation

Founded in 1998 and based in Dallas, Texas, Istation (Imagination Station) has become one of the nation's leading providers of richly animated, game-like educational technology. Istation was named one of 2019's "EdTech 1000: Companies Transforming Education" by EdTech Digest and has received numerous national educational technology awards. The Istation program puts more instructional time in the classroom through small-group and collaborative instruction. Istation's innovative reading, math and Spanish programs immerse students in an engaging and interactive environment and inspire them to learn. Additionally, administrators and educators can use Istation to easily track the progress of their students, schools and classrooms. Istation now serves more than 4 million students throughout the United States and in several other countries.

About Boulder Learning Inc.

Established in 2016, Boulder Learning is a learning technology company that is a pioneer in children's speech recognition technologies. Boulder Learning has been the recipient of over \$25M in research funding from the U.S. Department of Education, The Institute for Education Sciences and other agencies, and the company has successfully created and disseminated research-based programs such as Istation's oral reading fluency assessment in schools across the U.S. Through government-sponsored research and private investment, Boulder Learning is continuing to develop technologies for conversational intelligent tutors and automatic assessment systems. Virtual tutors for reading and science have been developed and rigorously evaluated in real-world conditions. Several of the systems are being readied for commercial development. Ancillary marketing and sales for HM will be provided by Sunburst Digital, a value added reseller currently marketing and selling HELP Math 1.5. Sunburst Digital is the successor company to Educational Resources and has been in the education business for over 30 years.

US K-12 Education Market: Organization & Size

We reviewed the target market in the US. States are responsible for the education of their residents. These states have 13,588 school districts. According to statistics compiled by various agencies and organizations (e.g., *(National Center for Education Statistics (NCES) Home Page, part of the U.S. Department of Education, n.d.)*; Center for Education Reform, *(K-12 Facts, n.d.)*), there are over 132,181 operational public elementary and secondary schools in the U.S. states and DC. Approximately 70% of students are in elementary school (K-8) and 30% in secondary school. There are about 75,000 elementary schools (including schools that combine elementary and secondary education). All together there are over 49.5 million public school students and 3.1 million teachers. It is possible to make rough estimates of market size from these numbers. For example, if every elementary school (75,000) in the US bought one of our programs for \$1000, revenue would be \$75 million. If every school bought one HELP program for each grade level, K-8, revenue would increase nine-fold, to \$675 million. This is, very roughly, the potential market for a K-8 HELP Math 2.0+R product line. A 1% share of this market is \$6.75 million annually. The global online education market is expected to reach approximately USD 132.98 billion by 2023, growing at a CAGR of 28.55% from 2017 to 2023. As such, we believe it is possible to increase revenues of HELP Math 2.0+R to over \$7 million within 2 years of release of HELP Math 2.0. We project sales to reach \$28 million after 5 years.

2. Project Time and Overview

The development of HM with HTML5 is based on an iterative, design-based development process (Thompson et al., 2004). In this type of design, as products and processes are being developed, they will be tested for feedback to developers. This in turn feeds improvements to the product which is continuously tested. In this case, the development will be the recoding of HM modules with HTML5. Initial testing will be complete with feedback about dysfunctions or "bugs" across multiple browsers. This will include using PC's, MAC's, iOS 13.1, Android KitKat, and Android 10. On each platform we will test HM lessons using Google Chrome, Mozilla Firefox, Opera, Microsoft Edge, Apple Safari, Vivaldi, Brave, and Tor. As a team we will build a compatibility table with every combination of browser and module. This will be followed with testing with preservice teachers at the University of Nebraska Lincoln. Table 3 shows a timeline below. Following the Table, the activities for each of the 8 months of the project are described.

Table 3. Timeline of Work Plan								
Tasks	Sep-20		Oct-20		Nov-20		Dec-20	
	Wk 1	Wk 2	Wk 1	Wk 2	Wk 1	Wk 2	Wk 1	Wk 2
HTML5 Coding	X	X	X	X	X	X	X	X
Team Testing			X	X	X	X	X	X
Teacher Testing								
Feedback				X	X	X	X	X
Surveys								
Focus Group Interviews								
Individual Interviews								

Tasks	Jan-21		Feb-21		Mar-21		Apr-21	
	Wk 1	Wk 2	Wk 1	Wk 2	Wk 1	Wk 2	Wk 1	Wk 2
HTML5 Coding	X					X	X	X
Teacher Testing			X	X	X	X		
Feedback		X		X	X	X	X	X
Surveys						X	X	
Focus Group Interviews							X	
Individual Interviews							X	X

3. Phase 1 Work Plan

a. Development of the prototype

Month 1: The team will hire specialists to begin recoding HM modules with HTML5. This will be an ongoing process that must start as soon as possible.

Month 2: The team will develop a protocol for reporting bugs to programmers based on their testing of the modules as they are developed with HTML5. Based on the initial recoding of HP using HTML5 we will test the effectiveness of HP using a variety of browsers and platforms. The five primary team members will complete a set of 32 HM lessons stratified across browser types.

Month 3: Just as in Month 2, the primary project team will continue testing HM modules across modules. They will complete another 32 HM lessons, reporting bugs to the programmers.

Month 4: This will entail the same process as month 3.

Month 5: We will finalize HTML5 coding and collect feedback through interviews with and surveys from the interviewers. We will obtain any feedback from them about the process by which debugging occurs such that alpha testing in the upcoming months with teachers proceeds smoothly.

Month 6: Preservice teachers at the University of Nebraska-Lincoln will begin testing HM modules. They will provide feedback to the programmers and project team as they test the HM modules using various browsers. Specifically, we will recruit 32 preservice teachers to test 2 modules per day for 2 days: 4 modules total. They will use various browsers to do so such that all HM modules are tested across every browser available.

Month 7: As in Month 6, preservice teachers will test HM modules using various browsers. Specifically, 32 additional preservice teachers will test 2 modules per day for 2 days: 4 modules total. They will again report findings to the programmers.

Month 8: Teachers will be provided with a survey about their experiences being alpha testers. Based on typical and atypical survey responses (Stake et al., 1994), teachers will be selected for focus group interviews. Based on selected focus group interview participation, teachers will be provided semistructured interviews about their experiences using HM and how it can be improved for broader teacher use.

b. Pilot Research at the End of Phase I

The pilot research, based on how this grant work is structured, is built into the development itself. In fact, as HM modules are recoded using HTML5 and tested first by the primary research team members and then by preservice teachers. The feedback provided throughout the project by the research team and teachers to the coders will result in HM modules that run using HTML5. Only exploratory research will be conducted based on the focus group and individual interviews during Month 8 with teachers. One primary research question will be explored: *How well did HTML5 allow the HM modules to function, if at all?*

We presume based on the time of investigation that every module will have finally worked across browsers and the primary question will be about how well HTML5 runs HM modules. Feedback from the focus group and individual interviews will be coded for themes (Erickson 1986). While there is a great deal more detail that can be provided about the type and nature of qualitative coding that will be conducted details about this are omitted due to space considerations. In sum, this will be a test of basic usability. It will support the commercialization plan for HM and the availability of it on platforms and browsers that broadly use HTML5.

c. Potential problems

Potential problems we anticipate exist in three primary areas. First, in the process of coding, the task may be overwhelming. If this is the case, we will simply hire additional programmers and have study personnel help in the coding. Second, the communication protocol between the project team, coders, with meetings, memos, emails, etc. will have to be carefully managed such that miscommunication or misinformation does not accidentally cause dysfunction. Third, during the testing of the product with teachers, if not enough teachers enroll to be testers that could be a problematic. If that is the case, we will incentivize participation with greater access (if not free access) to all HM products, and increase advertising outside colleges of education to include other pre-professionals. Overall, we do not anticipate many nonlogistical problems with the project, mostly because the proposal is largely logistical. It will make claims about the effectiveness of using HTML5 with HM, and be critical to the future of HM as a product that helps teachers and struggling students.

4. Project Team – Biographical Summary of Each Key Project Team Member

Brandon Holding, PhD, has worked, and is published, as a methodologist, statistician, and psychometrician. He has worked as methodologist on four NSF-funded grants, one IES-funded efficacy study, published one book on measurement development, and co-authored a handbook chapter on design research. Additionally, Dr. Holding is or has externally evaluated social science projects, has worked as a game developer for Google and the Hispanic Research Center at Arizona State University, as a policy consultant for the Lincoln Police Department and American Civil Liberties Union to detect patterns of bias in police officer behaviors with at-risk populations, the Arizona Department of Juvenile Detention Services in a study that helped determine which evaluation procedures corresponded with supports to reduce rates of recidivism, the Wellbeing

Initiatives, a center for improved mental health services for at-risk individuals, and a conglomerate of medical researchers with Dr. Susan Haag, Scientific Project Manager at CRESMET and AMTA at Arizona State University. They plan to establish evidence for advanced screening procedures for pancreatic cancer and possible clinical drug trials in junction with John Hopkins University. He will work as project leader and manager.

John Ramo M.Ed., has over 35 years experience in education entrepreneurship, educational technology and distance learning. Mr. Ramo founded and built a successful Washington D.C. educational software company, and completed a successful IPO on the NASDAQ. Recently, Mr. Ramo completed a successful merger of Digital Directions International (DDI) and Boulder Language Technologies to form Boulder Learning Inc. Previously with DDI, he formed a public private partnership with the University of Colorado, Rocky Mountain PBS to research and develop the award winning "HELP Math Program", the first research proven digital supplemental Math curriculum for Hispanic ELL and other struggling students under a major grants from the U.S. Department of Education and other foundations and private investment. Earlier in his career Mr. Ramo conceptualized and produced the first interactive multimedia music instruction software for the CD-I platform, The Private Lessons Series (Classical Guitar, Jazz Guitar, Rock Guitar, Piano), for Philips Interactive Media and he is an award-winning producer of educational multimedia software for other clients such as The Smithsonian Institute, Simon & Schuster, American Association for the Advancement of Science for whom he led the team that developed the first multimedia encyclopedia of the environment "Earth Explorer" for middle school students which was distributed by Apple. He has also worked for America On-Line as an Executive Producer, and on multi-media projects with names such as Jean Michel Cousteau John Denver, Jodie Foster and others. Mr. Ramo is a former Special Ed Teacher and has worked with children with special needs ranging from learning disabled, autism spectrum, to cognitive and behavioral disorders. He holds a Masters in Educational Psychology; fellowship awarded from American University and graduated with honors. He will work with the PI on project management.

Linda Hardman, a Learning Architect, is an award-winning K-12 math expert with more than 25 years in educational publishing. Linda has been successful in developing award-winning competitive products in mathematics such as CODiE award-winning *HELP Math*, Scholastic's *MATH 180* (a blended learning curriculum of print and online lessons/assessments and includes whole group, small group and individualized instruction (*Math 180 Program | 5-12 Math Intervention Curriculum | Houghton Mifflin Harcourt*, n.d.). Imagine Learning's *Imagine Math* (Think Through Math: 2012, 2015 CODiE awards for Best Mathematics Instructional Solution; an adaptive online language-rich grades 3 - Geometry math program that's personalized by the Quantile Framework, scaffolded, and provides on-demand tutoring by certified math teachers (*Imagine Math 3-8 | ImagineLearning*, n.d.), Harcourt's *Summer Studio Math*, and Prentice Hall's 6-12 math product arrays which also includes CODiE award-winning Prentice Hall's *Secondary Mathlab Toolkit*. Linda is also the lead author of *SAT Math Level 2 Subject Test Crash Course*, which has received raving reviews. Linda has extensive e-Learning, ELL, Intervention, and Special Education product development experience, along with extensive assessment development experience with high-stakes tests, such as TAKS, FCAT, SAT, Algebra EOC, GRE, and GMAT. She will oversee the overall process of collecting and collating feedback.

Megan Schmeister MA – With over 15 years in education, Megan has spent the last ten as Product Manager for HELP Math. Previous to working directly for HELP Math, Megan was part of two research grants, a US Department of Education Ready to Teach grant and a US Department of

Education IES Special Education grant. As part of that work, she has several publications, including an award for Article of the Year from Assessment for Effective Intervention. During the last ten years, as HELP Math's Product Manager, she has worked closely with teachers as they implement the program with their students. She will oversee the changes to HM that needs to be reflected in the coding.

Venkatesh Venkataramani, Venkatesh V. developed several educational technologies in his former role as Director of MVS Softech, a leading educational international technology company. As an industry leader in Corporate eLearning, interactive multimedia, and learning technologies, Venkatesh V. is committed to providing clients with cost-effective solutions utilizing various learning technologies. He has extensive research and development experience in hypermedia authoring, advanced scientific visualization, database management systems, and computer simulation. He will oversee the coding of HM lessons with HTML5.

5. External Evaluation - ReNU Evaluation narrative

The evaluation will be directed by Dr. Trish Wonch Hill, Research Coordinator of the Methodology and Evaluation Research Core Facility. Dr. Hill and her team of evaluation faculty and staff are currently evaluating dozens of grants focused on STEM education. Formative evaluation feedback will be presented to the program team monthly through the duration of the grant. This will provide information needed to most effectively meet the proposed goal and objectives. The impact of the activities relative to the program's goal and objectives will be assessed in the summative evaluation. The formative and summative evaluations will rely on a combination of qualitative and quantitative data.

To assess usability of HELP Math, MERC will assist in recruiting, consenting, tracking, and providing incentives to UNL pre-service teachers who elect to participate in testing the usability of up to four modules after the HELP Math HTML5 update. MERC evaluators will also recruit, host, and assist in the facilitation and provide transcript for 5 focus groups of HELP Math users, as well as facilitate follow-up interviews. MERC program staff will report feedback to the Project PI at Boulder Learning at each stage of the evaluation process, and will compile all results into a summative report at the culmination of the grant.

6. References

- Baluch, D. P., & Page Baluch, D. (2016). Jumpstarting STEM Careers. In *Forward to Professorship in STEM* (pp. 63–76). <https://doi.org/10.1016/b978-0-12-800855-3.00004-0>
- Behr, M. J., Harel, G., Post, T. R., & Lesh, R. (1992). *Rational number, ratio, and proportion*. <https://psycnet.apa.org/record/1992-97586-014>
- Beyranevand, M. L. (2014). The Different Representations of Rational Numbers. In *Mathematics Teaching in the Middle School* (Vol. 19, Issue 6, p. 382). <https://doi.org/10.5951/mathteachmiddscho.19.6.0382>
- Bond, E. (2014). *Childhood, Mobile Technologies and Everyday Experiences: Changing Technologies=Changing Childhoods?* Palgrave Macmillan, London.
- Brünken, R., Steinbacher, S., Plass, J. L., & Leutner, D. (2002). Assessment of cognitive load in multimedia learning using dual-task methodology. *Experimental Psychology*, 49(2), 109–119.

- CCSS. (2010). *Common core state standards for English language arts & literacy in history/social studies, science, and math*. Common Core State Standards Initiative. http://www.corestandards.org/assets/CCSSI_ELA%20Standards.pdf
- Collier, V. P. (1987). Age and Rate of Acquisition of Second Language for Academic Purposes. *TESOL Quarterly*, 21(4), 617–641.
- Council of the Great City Schools. (2014). *A Framework for Raising Expectations and Instructional Rigor for English Language Learners*. ERIC Clearinghouse.
- Cummins, J. (1979). Linguistic Interdependence and the Educational Development of Bilingual Children. *Review of Educational Research*, 49(2), 222–251.
- Cummins, J. (1981). The role of primary language development in promoting educational success for language minority students. *Schooling and Language Minority Students. A Theoretical Framework*. <https://ci.nii.ac.jp/naid/10029777893/>
- David, M. (2012). Building Websites With HTML5 to Work With Mobile Phones. In *HTML5 Mobile Websites* (pp. 3–53). <https://doi.org/10.1016/b978-0-240-81813-9.00007-0>
- de Jong, T., Linn, M. C., & Zacharia, Z. C. (2013). Physical and virtual laboratories in science and engineering education. *Science*, 340(6130), 305–308.
- Drew Polly, & Chandra Orrill. (2012). CCSSM: Examining the Critical Areas in Grades 5 and 6. *Teaching Children Mathematics*, 18(9), 566–573.
- Echevarria, J., & Short, D. (1999). The Sheltered Instruction Observation Protocol: A tool for teacher-researcher collaboration and professional development. *ERIC Digest EDO*, 99. <http://esol2.pbworks.com/f/The%20Sheltered%20Instruction%20Observation%20Protocol.doc>
- Haag, J. (2011). From elearning to mlearning: the effectiveness of mobile course delivery. *The Interservice/Industry Training, Simulation & Education Conference (IITSEC)*, 2011. <http://www.academia.edu/download/32546183/11053-paper.pdf>
- Halle, T., Castro, D., Franco, X., McSwiggan, M., Hair, E., & Wandner, L. (2011). The role of early care and education in the development of young Latino dual language learners. *Latina and Latino Children and Mental Health*, 1, 63–90.
- Heward, Y. (2009). Setting a poor example. In *Paediatric Nursing* (Vol. 21, Issue 3, pp. 8–8). <https://doi.org/10.7748/paed.21.3.8.s9>
- K-12 Facts. (n.d.). The Center for Education Reform. Retrieved February 27, 2020, from <https://www.edreform.com/2012/04/k-12-facts/>
- Krashen, S. D. (1992). *Fundamentals of language education*. Laredo Publishing Company.
- Lemke, J. (1998). Multiplying meaning: Visual and verbal semiotics in scientific text. In J. R. Martin & R. Veel (Eds.), *Reading Science* (pp. 87–113). Routledge.
- Marzano, R. J. (2004). *Building Background Knowledge for Academic Achievement: Research on What Works in Schools*. ASCD.
- Mayer, R. (2005). *The Cambridge Handbook of Multimedia Learning* (R. Mayer (Ed.); pp. 169–182). Cambridge University Press.

- Mayer, R. E. (2001). *Multimedia Learning*. Cambridge University Press.
- Mayer, R. E. (2005). Introduction to multimedia learning. *The Cambridge Handbook of Multimedia Learning*, 2, 1–24.
- National Center for Education Statistics (NCES) Home Page, part of the U.S. Department of Education. (n.d.). Retrieved February 27, 2020, from <https://nces.ed.gov/>
- Olympiou, Zacharia, & deJong. (2013). Making the invisible visible: Enhancing students' conceptual understanding by introducing representations of abstract objects in a simulation. *Instructional Science*, 41.
- Partnership for Assessment of Readiness for College and Careers - SAGE Research Methods. (n.d.). <https://doi.org/10.4135/9781506326139.n507>
- Resnick, M. L. (2005). Human factors in digital rights management for consumer products. In *PsycEXTRA Dataset*. <https://doi.org/10.1037/e577472012-012>
- Score, C. in R. (n.d.). *Student Increase on Post-test by Grade Level*. <https://www.helpprogram.net/DealerDocs/HELP%20Math%20Evaluation%20White%20Paper%205-13.pdf>
- Smarter Balanced Assessment Consortium. (2010). *Smarter Balanced Assessment Consortium: Theory of Action: An Excerpt from the Smarter Balanced Race to the Top Application*. ERIC Clearinghouse.
- Stake, R. E., Denzin, N. K., Lincoln, Y. S., & Oaks, C. A. (1994). Case studies. In *Handbook of Qualitative Research* Pp Thousand Sage Publications Inc, 236–247 SRC – GoogleScholar.
- Sweller, J., & Chandler, P. (1991). Evidence for Cognitive Load Theory. In *Cognition and Instruction* (Vol. 8, Issue 4, pp. 351–362). https://doi.org/10.1207/s1532690xc0804_5
- Thompson, T., Heer, D., Brown, S., Traylor, R., & Fiez, T. S. (2004). Educational design, evaluation, development of platforms for learning. *34th Annual Frontiers in Education, 2004. FIE 2004.*, T3E/1–T3E/4 Vol. 1.
- Viesca, K. M., Joseph, N. M., & Commins, N. L. (2019). A Framework for Improving the Teaching of Mathematics to Bi/Multilingual Learners. In *Teaching the Content Areas to English Language Learners in Secondary Schools* (pp. 135–150). https://doi.org/10.1007/978-3-030-02245-7_9
- Vygotsky, L. S., & Cambridge, M. A. (1978). *Mind in society: The development of higher psychological processes*.
- Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry, and Allied Disciplines*, 17(2), 89–100.
- Wouters, P., Paas, F., & Van Merriënboer, J. (2005). Observational learning from multimedia based expert models: The relation between modality, pacing and segmentation. *The European Conference for Research on Learning and Instruction, Nicosia, Cyprus*. <http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.87.2527&rep=rep1&type=pdf>
- WWC Review of the Report “Help with English Language Proficiency ‘HELP’ Program Evaluation of Sheltered Instruction Multimedia Lessons.” (n.d.).

Appendix A

No Prior Fast-Track or Phase II SBIR awards to report.

Appendix B

Letter of Support from University of Nebraska-Lincoln

February 28, 2020

VIA EMAIL
NUgrant ID# 133910

Brandon Holding, Ph.D
269 Sam Grange Ct.,
Carbondale, CO, USA 81623

Re: UNL Institutional Letter of Commitment

Dear Dr. Holding:

The Board of Regents of the University of Nebraska, for the University of Nebraska-Lincoln (UNL) is pleased to submit a proposal to your organization on behalf of Patricia Wonch Hill, for the project titled "Evaluating HELP Math HTLM5 Update."

The total request is for \$59,039 for the proposed period 09/01/2020 - 05/31/2021. Our EIN # is 47-0049123, our DUNS # is 55-545-6995, and our Congressional District is NE-001. Our participation is administratively approved for submission on behalf of the Board of Regents by the appropriate University officials.

Questions regarding the technical aspects of this project should be directed to Dr. Wonch Hill at (402) 472-7646 or phill3@unl.edu. Administrative questions should be directed to Kacey Nelkin Pedersen at (402) 472-6174 or knelkinpedersen2@unl.edu. We look forward to our involvement in this project.

Sincerely,



Kacey Nelkin Pedersen, CRA
Grants Coordinator III
Office of Sponsored Programs

Appendix C

They are included below. The remainder of this page is intentionally left blank.

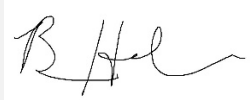
(1) Certifications (1-page)

Each individual on the project team must certify the following statement by providing the information below:

"I hereby certify that this information is accurate to the best of knowledge and belief."

Individual's Name (TYPED): Brandon Holding

Written signature: _____



Date: 3/1/20

Each employee of the offeror with the authority to bind the offeror must certify the following statement by providing the information below:

"I hereby certify that Boulder Learning, Inc. has verified the foregoing information that is accurate to the best of our knowledge and belief."

Individual's Name (TYPED): Brandon Holding

Individual's Title (TYPED): PI / Methodologist

Written signature: _____



Date: 3/1/20

BRANDON HELDING
Boulder Learning, Inc.
bhelding@boulderlearning.com

EDUCATION

Arizona State University	Elementary Education	B.A., 2005
Arizona State University	Curriculum and Instruction (Math Education)	Ph.D., 2010

PROFESSIONAL EXPERIENCE

2009 – Present	Statistician/Methodologist, Boulder Language Technologies
2007 – Present	Statistical consulting for Dr. Clark and Dr. Mathur, special education grant investigating factors contributing to recidivism (ongoing indefinitely)
2007 – Present	Methodological consulting for Dr. Bahadır Yanik, Turkey International Education and Career Center, proposal, Anatolyo University, Turkish Ministry of Education, Turkey (ongoing indefinitely)
2009 – 2010	Postdoctoral research associate, “Moving from Arithmetic to Algebra in Murphy School District: Multilevel Insights” (Grant No. 0634103)
2008 – 2010	Gaming consultant, Hispanic Research Center, Arizona State University
2007 – 2009	Statistical analyst, Math and Science Partnership, data
2007 – 2008	Statistical consulting for Dr. Oakes, special education grant

SELECTED PUBLICATIONS

1. Lewis, E.B., Baker, D.R., & Holding, B.A. (2015). Science teaching reform through professional development: Teachers’ use of a scientific classroom discourse community model. *Science Education*, 99(5), 896-931.
2. Holding, B. (2010). *Measurement in the service of mathematics education research: The case of Cognitively Guided Instruction*. LAP Lambert Academic Publishing: Saarbrücken, Germany.
3. Holding, B., Megowan-Romanowicz, M. C., Ganesh, T., & Fang, S. (2009). Interdisciplinary modeling instruction: Helping fifth graders learn about levers. In R. Lesh, P. Galbraith, C. Haines, & A. Hurford (Eds.), *Modeling Students’ Mathematical Modeling Competencies: ICTMA 13*. New York, NY: Springer Science + Business Media, Inc.
4. Sloane, F., Holding B. & Kelly, A. (2008). Interrupted time series designs for design research in mathematics education. In A. Kelly & R. Lesh (Eds.). *The handbook of design research in mathematics, science and technology education*. New York, NY: Taylor Francis.
5. Sloane, F., Holding, B. & Kelly, A. (2008). Modeling student development in science education: An introduction to growth modeling. In K. Tobin, & M. Wolfe (Eds.). *The World of Science Education: North America*.

SYNERGISTIC ACTIVITIES

1. Middleton, J., Holding, B., Carruthers, C., Megowan-Romanowicz, C., Kim, A., Kim, Y., Yanik, B., Flores, A., & Atkinson, R. (2008). *A Longitudinal Study of Rational Number Understanding in the Middle-Grades: Final Report to the National Science Foundation*. Center for Research on Education in Science, Mathematics, Engineering, and Technology.
2. Cole, R., & Holding, B. (2016). Teachers’ and Children’s Experiences with MindStars Books. *Final report to the National Science Foundation*, Boulder Learning, Inc.
3. Cole, R. & Holding, B. (2015). Teachers’ and Children’s Experiences with MindStars Books. *Report submitted to Boulder Valley School District*, Boulder Learning, Inc.
4. External Evaluation Consultant: Dr. Ganesh, WATER Science Professional Development.
5. Consulting Methodologist: Dr. Haag, Power analyses for Clinical trials for upcoming screening processes for pancreatic cancer – potential external evaluation.
6. Consulting Statistician: American Civil Liberties Union, Working on two projects: 1) examine bias in Lincoln Police Department detainment, and 2) examine potential differential rates of revoked bail for underrepresented or at-risk groups control for demographic information..
7. External Evaluator: STEMteachersNYC. Grant providing Professional Development.

Appendix C

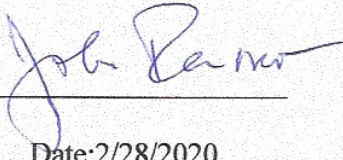
(1) Certifications (1-page)

Each individual on the project team must certify the following statement by providing the information below:

"I hereby certify that this information is accurate to the best of knowledge and belief."

Individual's Name: John Ramo

Written signature:

A handwritten signature in blue ink, appearing to read "John Ramo", is written over a horizontal line.

Date: 2/28/2020

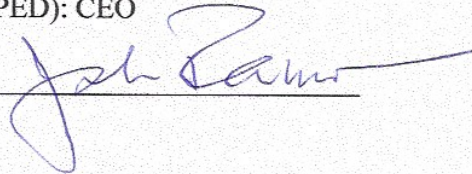
Each employee of the offeror with the authority to bind the offeror must certify the following statement by providing the information below:

"I hereby certify that (Boulder Learning Inc.) has verified the foregoing information that is accurate to the best of our knowledge and belief."

Individual's Name (TYPED): John Ramo

Individual's Title (TYPED): CEO

Written signature:

A handwritten signature in blue ink, appearing to read "John Ramo", is written over a horizontal line.

Date:

Vitae

John Ramo M.Ed.,

John Ramo has over 35 years of experience in education entrepreneurship, educational technology and digital learning. Mr. Ramo founded and built a successful Washington D.C. educational software company and completed a successful IPO on the NASDAQ. In 2017, Mr. Ramo completed a successful merger of Digital Directions International (DDI) and Boulder Language Technologies to form Boulder Learning Inc. Previously with DDI, he formed a public private partnership with the University of Colorado, Rocky Mountain PBS to research and develop the award winning "HELP Math Program", the first research proven digital supplemental Math curriculum for Hispanic ELL and other struggling students under a major grants from the U.S. Department of Education and other foundations and private investment. Earlier in his career Mr. Ramo conceptualized and produced the first interactive multimedia music instruction software for the CD-I platform, The Private Lessons Series (Classical Guitar, Jazz Guitar, Rock Guitar, Piano), for Philips Interactive Media and he is an award-winning producer of educational multimedia software for other clients such as The Smithsonian Institute, Simon & Schuster, American Association for the Advancement of Science for whom he led the team that developed the first multimedia encyclopedia of the environment "Earth Explorer" for middle school students which was distributed by Apple. He has also worked for America On-Line as an Executive Producer, and on multi-media projects with names such as Jean Michel Cousteau John Denver, Jodie Foster and others. Mr. Ramo is a former Special Ed Teacher and has worked with Special Needs children ranging from learning disabled, autism spectrum, to cognitive and behavioral disorders. He holds a Master's in educational psychology; fellowship awarded from The American University and graduated with honors. Mr. Ramo's projects have won numerous awards in radio, film and digital education.

Appendix C

(1) Certifications (1-page)

Each individual on the project team must certify the following statement by providing the information below:

"I hereby certify that this information is accurate to the best of knowledge and belief."

Individual's Name (TYPED): Linda A. Hardman

Written signature: Linda A. Hardman

Date: February 29, 2020

Each employee of the offeror with the authority to bind the offeror must certify the following statement by providing the information below:

"I hereby certify that (FILL IN THE COMPANY NAME) has verified the foregoing information that is accurate to the best of our knowledge and belief."

Individual's Name (TYPED): Linda A. Hardman

Individual's Title (TYPED): President, Linda A. Hardman Consulting Inc.

Written signature: Linda A. Hardman

Date: February 29, 2020

LINDA A. HARDMAN

6800 Canon Wren Drive Austin, TX 78746
512-330-9670 512-299-2017 (cell) jhardman@austin.rr.com

Publishing, Assessment, and Learning Technologies Executive with proven leadership who has been successful in developing award-winning/competitive products in mathematics, science, language arts, and social studies. Has produced highly comprehensive products (core curriculum, assessment/testing, textbooks, online learning courseware, technology supplementals, iPad and apps, professional development training, and learning aides) while addressing state/CCSS objectives/benchmarks and high-stakes tests. Has had personal responsibility for:

- **Product and Assessment Development:**
 - Project Management
 - Editorial
 - Content Writing: Teacher and Student Resources
 - Assessment Writing and Development
 - Meeting budgetary constraints and revenue targets
 - Delivering products on-time
- **Lead Author:** *SAT Math Level 2 Subject Test Crash Course*
- **Extensive e-Learning,** ELL, Intervention, and Special Education Product Development
- **Assessment Development:** high-stakes tests, such as TAKS, FCAT, SAT, Algebra EOC, GRE, and GMAT

Award-winning Products: *HELP Math:* CODiE Award 2009 and 2010: **Best Instructional Solution for Special Needs Students** and CODiE Awards Finalist 2009: **Best Mathematics Instructional Solution; Think Through Math (Imagine Math):** 2012, 2014, 2015 CODiE awards for Best Mathematics Instructional Solution and a finalist in 2016; *Harcourt's Summer Studio Math (grades 1-6)*, Teacher's Choice Award; Prentice Hall's *Secondary Mathlab Toolkit* for middle school and high school (CODiE Award)

Experience

Imagine Learning, Provo, UT

August 2016 – December 2018

Executive Director Instructional Design, Mathematics

Directed the design, development, and planning of all mathematics products which includes the re-design of acquired mathematics products to increase students' conceptual and procedural understanding of mathematics concepts and skills for grades PK - Geometry. Developed a collaboration with **The Dana Center**.

Linda A. Hardman Consulting Inc. (principal), Austin, TX

2000 — Present

Certified Texas Historically Underutilized Business (HUB)

Provided consultant services in project management, product and assessment development, and training for technology/web applications, eLearning, courseware, and textbook products, all of which included working in learning/content management systems. Clients include: Scholastic Education/Houghton Mifflin Harcourt, Redbird Learning (now McGraw-Hill), Digital Directions International (now Boulder Learning), Compass Learning (now Edgenuity Inc.), Measured Progress, Pearson Assessment, Houghton Mifflin Harcourt, EdTech Games, and Ellevation.

Scholastic Education, New York, NY

June 2012 – August 2013

Director Math Publishing. Led the development of all math publishing products, which included *MATH 180*.

Harcourt Achieve, Austin, TX

2004 – 2005

Vice President Product Development Mathematics (*Saxon Math, Steck-Vaughn, and Rigby*).

Bell & Howell Information & Learning (ProQuest), Ann Arbor, MI

Spring 1999 – Fall 2000

Vice President (Science, Technology, and Mathematics & Medical) and Publisher

Cogito Learning Media, New York City, and San Francisco, CA

1998 – 1999

Vice President, Business Development & Special Projects

Pearson Education, Prentice Hall School Division, NJ

1994 – 1998

Vice President, Educational Technology

Appendix C

(1) Certifications (1-page)

Each individual on the project team must certify the following statement by providing the information below:

"I hereby certify that this information is accurate to the best of knowledge and belief."

Individual's Name (TYPED): Megan Schmeister

Written signature: 

Date: March 1, 2020

Each employee of the offeror with the authority to bind the offeror must certify the following statement by providing the information below:

"I hereby certify that (Boulder Learning Inc.) has verified the foregoing information that is accurate to the best of our knowledge and belief."

Individual's Name (TYPED): Megan Schmeister

Individual's Title (TYPED): Product Manager

Written signature: 

Date: March 1, 2020

Megan Schmeister

EDUCATION:

M.A. in Special Education: December 2006
University of Colorado – Colorado Springs, CO

B.A. in Psychology: December 1998
University of Colorado – Colorado Springs, CO

EXPERIENCE:

Product Manager / Head of Teacher Support

Boulder Learning / Digital Directions International
August 2010 – present

Professional Research Assistant

University of Colorado - Colorado Springs, College of Education
August 2009 – May 2010

- U.S. Department of Education, Institute of Education Statistics (IES), Special Education Research Grant
- The Math Learning Companion: An Individualized Intervention for Students with Math Learning Disabilities

Professional Research Assistant

University of Colorado - Colorado Springs, College of Education
June 2006 – May 2009

- U.S. Department of Education, Ready to Teach grant
- Help with English Language Proficiency (HELP) Program

Adjunct Professor

University of Colorado - Colorado Springs, College of Education, Special Education
January 2008 – May 2008

Resource Teacher

Colorado Springs School District 11, Special Education
August 2005 – May 2006

Job Coach / Educational Assistant

Colorado Springs School District 11, Special Education / Alternative Cooperative Education (ACE)
August 2004 – August 2005

Substitute Teacher

Colorado Springs School District 11
October 2003 – August 2004

PUBLICATIONS:

Crawford, L., Carpenter, D. M., Wilson, M. T., Schmeister, M., & McDonald, M. (2012). Testing the Relationship between Fidelity of Implementation and Student Outcomes in Math. *Assessment for Effective Intervention*, 37(4), 224-235.

Crawford, L., Schmeister, M., & Biggs, A. (2008). Impact of intense professional development on teachers' use of sheltered instruction with students who are English language learners. *Journal of Inservice Education*, 34(3), 327-342.

University of Colorado, Colorado Springs. (2006). *Freedom's Song: 100 Years of African American Struggle and Triumph*. Colorado Springs, Colorado: Association for the Study of African American Life and History (ASALH).

PRESENTATIONS:

Schmeister, M. (November 2011). HELP Implement for Response to Intervention (RtI) for ELL Math. Presentation at the National Council of Teachers of Mathematics (NCTM) Regional Conference & Exposition, Albuquerque, NM.

Crawford, L., Schmeister, M., & McDonald, M. (April 2009). Researching the effects of a supplemental mathematics curriculum for English language learners. Paper presented at session Promising Applied Research Practices for Second Language Learners at the 2009 American Educational Research Association (AERA) Annual Meeting - Disciplined Inquiry: Education Research in the Circle of Knowledge, San Diego, CA.

Freeman, B., & Schmeister, M. (October 2007). Help for English language proficiency (HELP) program. Presentation at the Annual Colorado Council of Teachers of Mathematics (CCTM) Conference, Denver, CO.

Crawford, L., Lawyer-Brook, D., Schmeister, M., McDonald, M., & Franco, L. (March 2007). Effects of a web-based mathematics program. Presentation at the 41st Annual Teachers of English to Speakers of Other Languages (TESOL) Convention and Exhibit, Seattle, WA.

AWARDS:

Article of the Year: April 2013 – Assessment for Effective Intervention

Appendix C

(1) Certifications (1-page)

Each individual on the project team must certify the following statement by providing the information below:

"I hereby certify that this information is accurate to the best of knowledge and belief."

Individual's Name (TYPED): VENKATESH VENKATARAMANI



Written signature: _____

Date:

Each employee of the offeror with the authority to bind the offeror must certify the following statement by providing the information below:

"I hereby certify that (FILL IN THE COMPANY NAME) has verified the foregoing information that is accurate to the best of our knowledge and belief."

Individual's Name (TYPED): VENKATESH VENKATARAMANI

Individual's Title (TYPED): INDEPENDENT CONSULTANT



Written signature: _____

Date:

Resume

V. VENKATESH

Email ID : mvssoftech@gmail.com

Education : B.Sc. Mathematics from Madras University
Higher Diploma in Software Engineering
From Aptech.

Software skills : Adobe Suite, HTML, HTML5, MS-Office, other multimedia
Tools, .NET

PERSONAL DETAILS

Name : V. Venkatesh

Date of birth : 21-02-1975

Nationality : Indian

Experience : **Multimedia Designer @** Pentamedia Graphics (1 Year) -1998-1999
Visual Designer @ SSI Limited – 1999 - 2001
Project Lead @ Til Studios -2001-2002
Director @ MVS Softech – 2002 – 2018
Independent Consultant – 2019 – till date

Declaration:

I hereby declare that the statement made in this application are true complete and correct to the best of my knowledge and belief.

Sincerely,

[V. VENKATESH]

Appendix D

There is nothing applicable to report.

Appendix E

PHASE I BUDGET INFORMATION (MAXIMUM OF \$200,000 FOR 8-MONTHS)

- Budget Narrative Justification for Phase I (limit 1-page for the narrative):
- Spreadsheet for Phase I (use the template below):

Price Category		HOURS		RATE	TOTAL
Labor					
Brandon Holding, Principle Investigator		574	X	\$20.83	\$11956.42
John Ramo, Co-Principle Investigator		574	X	\$20.83	\$11956.42
Linda Hardman, Co-Principle Investigator		574	X	\$20.83	\$11956.42
Megan Schmeister, Co-Principle Investigator		574	X	\$20.83	\$11956.42
Venkatesh V, Co-Principle Investigator		574	X	\$20.83	\$11956.42
Labor Overhead	Rate	0%			\$0
Fringe Benefits		0%			\$0
Total Direct Cost					\$59782.10
Other Direct Costs					
Travel					\$0
Copy/print/message					\$200.90
Technical Assistance (maximum of \$6,500) (See Appendix H)					\$0
Consultant					\$80000
Subcontractors(<i>no more than 33% of the total Phase I price</i>)					\$58,969
Total ODCs					\$140200
Total Direct Costs					\$59782.10
G&A	Rate	%			\$0
Total Cost					\$199169
Profit	Rate	%			\$0
TOTAL PRICE					\$199169

Funding Agency: NSF Discovery Research PreK-12 (DRK12)

PI Name: Martha Gregg

Project Title: TBD

Project Start Date: 6/1/2020

Project End Date: 5/31/2023

UNL Basic Request Budget			
	Person Months	Year 1	Total
<i>Senior Personnel</i>	Yr1		
Patricia Wonch Hill	3.00	17,116	17,116
Total Senior Personnel		17,116	17,116
<i>Other Personnel</i>	# of Ppl		
Post Docs	0	-	-
Other Professionals	0	-	-
Graduate Students	0.00	-	-
Undergraduate Students	0	-	-
Secretarial	0	-	-
Other	0	-	-
Total Other Personnel		-	-
Fringe Benefits		6,846	6,846
Total Salaries and Benefits		23,962	23,962
Equipment		-	-
Travel		-	-
Supplies		100	100
Subawards		-	-
Other		13,860	13,860
Total Other Direct Costs		13,960	13,960
Total Direct Costs		37,922	37,922
F&A Base	TDC	37,922	
F&A	55.5%	21,047	21,047
Total Request		58,969	58,969

MTDC Exclusions	Year 1	Total
Equipment	-	-
Tuition Remission	-	-
Subawards in excess of \$25K	-	-
Participant Support Costs	-	-
Rent	-	-
Alterations and Renovations	-	-
Total Exclusions	-	-

HELP Math Budget Justification – Boulder Learning Inc.

The budget for this proposal, while nearing the maximum allotted amount, provides modest compensation for the primary research staff. Each primary researcher will receive \$11,956.42 in total compensation for 8 months of work.

Copy and print services include electronic storage for copying files and printing documents as part of dissemination efforts. No other costs are included in this section.

The bulk of the Boulder Learning, Inc. budget (\$80,000) is allocated for a consulting computer scientists or team of computer scientists that will be charged with recoding the HM lessons with HTML5. Because this is the primary work of the grant, this is justified.

The University of Nebraska-Lincoln will oversee the project (as external evaluator) and provide preservice teachers for data collection. They will also oversee the focus group and individual interviews, data management, and verify Institutional Review Board (IRB) clearances. As a result, they are budgeted at \$58,969.

Of particular note in this budget is that there is no profit allocated for Boulder Learning, Inc. We expect the profit to originate from the updated HM lessons with HTML5. This will correspond with our commercialization plan as well as the general business model of the company itself.

HELP Math Budget Justification – MERC

Personnel

Dr. Hill (Evaluation Lead, 0.3 calendar months) will coordinate all data collection and will be responsible for writing reports and reporting back to the MATH Help leadership and reporting to the funding agency.

Other Direct Costs

MERC Evaluation Service Center

The Methodology and Evaluation Research Core Facility, a service center, will be utilized to carry out data collection. Program evaluation manager, and project associates time is estimated at a total of 120 hours to assist in data collection for five focus groups and follow-up interviews. They will assist in qualitative data analysis and reporting. Undergraduates will enter data and assist with interview transcription for approximately 60 total hours.

The Methodology and Evaluation Research Core Facility (MERC) is a core facility housed within the Social and Behavioral Sciences Research Consortium at the University of Nebraska-Lincoln. The mission of the Methodology and Evaluation Research Core Facility (MERC) is to provide state-of-the-art methodological support and services that promote and sustain excellence in the social and behavioral sciences. MERC provides consultation on evaluation designs and the implementation of social science research and evaluation for funded grants. MERC provides evaluation for dozens of NSF projects, serving a variety of evaluation roles to best fit the investigators' needs, including serving as an internal evaluator, an external evaluator, an evaluation consultant, a social science expert, or providing trainings to build evaluation capacity.

Study Subject Payments

Approximately 50 pre-service teachers from UNL will be recruited to conduct 4-6 hours of usability testing. Each participant will test 4 modules of HELP Math using two different browsers and will be payed \$50 upon completion. Additionally, 25 pre-service teachers will participate in 3-5 60-90-minute focus groups and will receive \$10 for their participation. Finally, follow-up interviews with 5 pre-service teachers will last 30-60 minutes will be conducted prior to finalization of the update to smooth out final updates to HELP Math.

MERC Supplies

Costs are estimated at \$100/year in Years 1-3 to purchase consumable materials and supplies for the proposed work, such as printing costs for papers and posters and supplies for data collection.

Facilities & Administrative Cost

UNL's federally negotiated rate is 55.5% of MTDC for organized research.

	MTDC Base	F&A Costs
Y1	37,922	21,047

Appendix F

Human Subjects Protections

EDAR 3452.224-71 Notice About Research Activities Involving Human Subjects (MAY 2011)

- (a) All procedures are complied with.
- (b) All definitions are accepted.
- (c) This project is exempt under 34 CFR 97.101(b) (2): “Research involving the use of educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures or observation of public behavior”. Applicably, neither exclusions to is exemption This project is exempt under 34 CFR 97.101(b) (2), (i) or (ii) apply.

All participants in this project will provide anonymized responses in the form of bug reports to the project team and coders directly. Anonymization will occur at the source of the data information. That is, when the forms are completed and submitted, at no point will the submitter’s name be included in that form. There will be no manipulation of a naturally occurring environment.

Interviews (focus group and individual) will not include any identification information and will only be transcribed with pseudonyms.

- (d) No children will be included in the study.
- (e) This project is exempt under 34 CFR 97.101(b) (2): “Research involving the use of educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures or observation of public behavior”. Applicably, neither exclusions to is exemption This project is exempt under 34 CFR 97.101(b) (2), (i) or (ii) apply.
- (f) All research activities and exemptions will be processed through the Institutional Review Board (IRB) at the University of Nebraska-Lincoln (UNL).

Appendix G

Appendix G: Department of Education Required Information (Forms)

NOTE: Offerors without all completed certifications as required in Appendices G. will not be reviewed.

G. 1 FAR 52.203-17 - Contractor Employee Whistleblower Rights and Requirement To Inform Employees of Whistleblower Rights. (Apr 2014)

(a) This contract and employees working on this contract will be subject to the whistleblower rights and remedies in the pilot program on Contractor employee whistleblower protections established at 41 U.S.C. 4712 by section 828 of the National Defense Authorization Act for Fiscal Year 2013 (Pub. L. 112-239) and FAR 3.908.

(b) The Contractor shall inform its employees in writing, in the predominant language of the workforce, of employee whistleblower rights and protections under 41 U.S.C. 4712, as described in section 3.908 of the Federal Acquisition Regulation.

(c) The Contractor shall insert the substance of this clause, including this paragraph (c), in all subcontracts over the simplified acquisition threshold.

Certification G.1

The offeror has read and understood the above clause in this section.

XYes ☐No

G.2 FAR 52.204-19 – Incorporation by Reference of Representations and Certifications. (Dec 2014)

The Contractor's representations and certifications, including those completed electronically via the System for Award Management (SAM), are incorporated by reference into the contract.

(End of clause)

Certification G.2

The offeror has read and understood the above clause in this section.

XYes ☐No

G. 3 FAR 52.204-7 -- System for Award Management. (Oct 2016)

(a) Definitions. As used in this provision—

“Electronic Funds Transfer (EFT) indicator” means a four-character suffix to the unique entity identifier. The suffix is assigned at the discretion of the commercial, nonprofit, or Government entity to establish additional System for Award Management records for identifying alternative EFT accounts (see subpart 32.11) for the same entity.

“Registered in the System for Award Management (SAM) database” means that—

- (1) The Offeror has entered all mandatory information, including the unique entity identifier and the EFT indicator, if applicable, the Commercial and Government Entity (CAGE) code, as well as data required by the Federal Funding Accountability and Transparency Act of 2006 (see subpart 4.14), into the SAM database;
- (2) The offeror has completed the Core, Assertions, and Representations and Certification, and Points of contact sections of the registration in the SAM database;
- (3) The Government has validated all mandatory data fields, to include validation of the Taxpayer Identification Number (TIN) with the Internal Revenue Service (IRS). The Offeror will be required to provide consent for TIN validation to the Government as a part of the SAM registration process.
- (4) The Government has marked the record “Active”.

“Unique entity identifier” means a number or other identifier used to identify a specific commercial, nonprofit, or Government entity. See www.sam.gov for the designated entity for establishing unique entity identifiers.

(b)

- (1) By submission of an Offer, the offeror acknowledges the requirement that a prospective awardee shall be registered in the SAM database prior to award, during performance, and through final payment of any contract, basic agreement, basic ordering agreement, or blanket purchasing agreement resulting from this solicitation.
- (2) The Offeror shall enter, in the block with its name and address on the cover page of its offer, the annotation “Unique Entity Identifier” followed by the unique entity identifier that

identifies the Offeror's name and address exactly as stated in the offer. The Offeror also shall enter its EFT indicator, if applicable. The unique entity identifier will be used by the Contracting Officer to verify that the Offeror is registered in the SAM database.

(c) If the Offeror does not have a unique entity identifier, it should contact the entity designated at www.sam.gov for establishment of the unique entity identifier directly to obtain one. The Offeror should be prepared to provide the following information:

- (1) Company legal business name.
- (2) Trade style, doing business, or other name by which your entity is commonly recognized.
- (3) Company physical street address, city, state and Zip Code.
- (4) Company mailing address, city, state and Zip Code (if separate from physical).
- (5) Company telephone number.
- (6) Date the company was started.
- (7) Number of employees at your location.
- (8) Chief executive officer/key manager.
- (9) Line of business (industry).
- (10) Company Headquarters name and address (reporting relationship within your entity).

(d) If the Offeror does not become registered in the SAM database in the time prescribed by the Contracting Officer, the Contracting Officer will proceed to award to the next otherwise successful registered Offeror.

(e) Processing time, which normally takes 48 hours, should be taken into consideration when registering. Offerors who are not registered should consider applying for registration immediately upon receipt of this solicitation.

(f) Offerors may obtain information on registration at <https://www.acquisition.gov>.

(End of Provision)

Will the offeror be registered in the SAM Database prior to award? ☒ Yes ☐ No

G.4 SBA Company Registry

All offerors to the SBIR program are required to register at the SBA Company Registry (<http://sbir.gov/registration>) prior to application submission. At the bottom of section G.2 of the proposal, offerors are required to attach a .pdf FILE of the SBIR.gov SBC Registration Control ID Form as proof of registration in this section of the proposal. **Proposals that do not include the .pdf file will be returned without review.** See below a picture of a blank SBIR.gov Registration Control ID Form that offerors will complete, save as a .pdf FILE, and include within the proposal.



SBIR.gov SBC Registration Control ID Form

Company Information				
SBC Control ID				
Company Name				
Address				
City		State	Zip	
TIN/EIN		DUNS		
Company URL				
Number of Employees				
Is this SBC majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms?				
What percentage (%) of the SBC is majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms?				

Notes: Completed registrations will receive a unique SBC Control ID and .pdf file. If previously registered, offerors are still required to attach proof of registration. The SBA Company Registry recommends verification with SAM, but a SAM account is not required to complete the registration. In order to be verified with SAM, your email address must match one of the contacts in SAM. If unsure if what is listed in SAM for your company, offerors may verify the information on the SAM site. Confirmation of a company's DUNS is necessary to verify an offeror's email address in SAM.

Follow these steps listed below to register and attach proof of registration to the proposal.

a. Navigate to the SBA Company Registry (<http://sbir.gov/registration>).

- b. If you are a previous SBIR awardee from any agency, search for your small business by Company Name, EIN/Tax ID, DUNS, or Existing SBIR Contract/Grant Number in the search fields provided. Identify your company and click “Proceed to Registration”.
- c. If you are a first time applicant, click the New to the SBIR Program (http://sbir.gov/firm_user_register) link on lower right of registry screen.
- d. Fill out the required information on the “Basic Information” and “Eligibility Statement” screens.
- e. Press “Complete Registration” on the lower right of the “Eligibility Statement” screen and follow all instructions.
- f. Download and save your SBA registry PDF locally. The name will be in the format of SBC_123456789.pdf, where SBC_123456789 (9 digit number) is your firm’s SBC Control ID.

Certification G.4

Attach the pdf. FILE for the SBIR.gov SBC Registration Control ID Form here: see note below attachment.



SBIR.gov SBC Registration

SBC Control ID:	SBC_000562607		
Company Name:	Boulder Language Technologies (BLT), Inc.		
Address:	2960 Center Green Ct Ste 200		
City:	Boulder		
State:	CO	Zip:	80301-5406
EIN (TIN):	208379918	DUNS:	791197622
Company URL:	www.bltek.com		
Number of Employees:			11
Is this SBC majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms?			No
What percentage (%) of the SBC is majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms?			0.00%

NOTE: The company name has changed to BOULDER LEARNING, INC.

G.5. SBIR Funding Agreement Certification

All small businesses are required to complete the below certification form in order to be selected for an SBIR award. This includes checking all of the boxes and having an authorized officer of the awardee sign and date the certification each time it is requested. Proposals that do not include the completed certification will be returned without review.

Certification G.5

Please read carefully the following certification statements. The Federal government relies on the information to determine whether the business is eligible for a Small Business Innovation Research (SBIR) Program award. A similar certification will be used to ensure continued compliance with specific program requirements during the life of the funding agreement. The definitions for the terms used in this certification are set forth in the Small Business Act, SBA regulations (13 C.F.R. Part 121), the SBIR Policy Directive and also any statutory and regulatory provisions referenced in those authorities.

If the funding agreement officer believes that the business may not meet certain eligibility requirements at the time of award, they are required to file a size protest with the U.S. Small Business Administration (SBA), who will determine eligibility. At that time, SBA will request further clarification and supporting documentation in order to assist in the verification of any of the information provided as part of a protest. If the funding agreement officer believes, after award, that the business is not meeting certain funding agreement requirements, the agency may request further clarification and supporting documentation in order to assist in the verification of any of the information provided.

Even if correct information has been included in other materials submitted to the Federal government, any action taken with respect to this certification does not affect the Government's right to pursue criminal, civil or administrative remedies for incorrect or incomplete information given in the certification. Each person signing this certification may be prosecuted if they have provided false information.

The undersigned has reviewed, verified and certifies that (all boxes must be checked):

(1) The business concern meets the ownership and control requirements set forth in 13 C.F.R. §121.702.

(2) ☒ Yes ☐ No

(2) If a corporation, all corporate documents (articles of incorporation and any amendments, articles of conversion, by-laws and amendments, shareholder meeting minutes showing director elections, shareholder meeting minutes showing officer elections, organizational meeting minutes, all issued stock certificates, stock ledger, buy-sell agreements, stock transfer agreements, voting agreements, and documents relating to stock options, including the right to

convert non-voting stock or debentures into voting stock) evidence that it meets the ownership and control requirements set forth in 13 C.F.R. §121.702.

XYes ☐No ☐N/A **Explain why N/A:**

(3) If a partnership, the partnership agreement evidences that it meets the ownership and control requirements set forth in 13 C.F.R. §121.702.

☐Yes ☐No **X**N/A **Explain why N/A: It is not a partnership.**

(4) If a limited liability company, the articles of organization and any amendments, and operating agreement and amendments, evidence that it meets the ownership and control requirements set forth in 13 C.F.R. §121.702.

☐Yes ☐No **X**N/A **Explain why N/A: It is not an LLC.**

(5) The birth certificates, naturalization papers, or passports show that any individuals it relies upon to meet the eligibility requirements are U.S. citizens or permanent resident aliens in the United States.

XYes ☐No ☐N/A **Explain why N/A:**

(6) It has no more than 500 employees, including the employees of its affiliates.

XYes ☐No

(7) SBA has not issued a size determination currently in effect finding that this business concern exceeds the 500 employee size standard.

XYes ☐No

(8) During the performance of the award, the principal investigator will spend more than one half of his/her time as an employee of the awardee or has requested and received a written deviation from this requirement from the funding agreement officer.

XYes ☐No ☐Deviation approved in writing by funding agreement officer: %

(9) All, essentially equivalent work, or a portion of the work proposed under this project (check the applicable line):

X Has not been submitted for funding by another Federal agency.

☐ Has been submitted for funding by another Federal agency but has not been funded under any other Federal grant, contract, subcontract or other transaction.

☐ A portion has been funded by another grant, contract, or subcontract as described in detail in the proposal and approved in writing by the funding agreement officer.

(10) During the performance of award, it will perform the applicable percentage of work unless a deviation from this requirement is approved in writing by the funding agreement officer (check the applicable line and fill in if needed):

XSBIR Phase I: at least two-thirds (66 2/3%) of the research.

☐ **Deviation approved in writing by the funding agreement officer: %**

(11) During performance of award, the research/research and development will be performed in the United States unless a deviation is approved in writing by the funding agreement officer.

XYes ☐ **No** ☐ **Waiver has been granted**

(12) During performance of award, the research/research and development will be performed at my facilities with my employees, except as otherwise indicated in the SBIR application and approved in the funding agreement.

XYes ☐ **No**

(13) It has registered itself on SBA's database as majority-owned by venture capital operating companies, hedge funds or private equity firms.

Yes **XNo**

☐ **N/A Explain why N/A:**

(14) It is a Covered Small Business Concern (a small business concern that: (a) was not majority- owned by multiple venture capital operating companies (VCOCs), hedge funds, or private equity firms on the date on which it submitted an application in response to an SBIR solicitation; and (b) on the date of the SBIR award, which is made more than 9 months after the closing date of the solicitation, is majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms).

☐ **Yes** **XNo**

X It will notify the Federal agency immediately if all or a portion of the work authorized and funded under this award is subsequently funded by another Federal agency.

X I understand that the information submitted may be given to Federal, State and local agencies for determining violations of law and other purposes.

XI am an officer of the business concern authorized to represent it and sign this certification on its behalf. By signing this certification, I am representing on my own

behalf, and on behalf of the business concern that the information provided in this certification, the application, and all other information submitted in connection with this application, is true and correct as of the date of submission. I acknowledge that any intentional or negligent misrepresentation of the information contained in this certification may result in criminal, civil or administrative sanctions, including but not limited to: (1) fines, restitution and/or imprisonment under 18 U.S.C. §1001; (2) treble damages and civil penalties under the False Claims Act (31 U.S.C. §3729 et seq.); (3) double damages and civil penalties under the Program Fraud Civil Remedies Act (31 U.S.C. §3801 et seq.); (4) civil recovery of award funds, (5) suspension and/or debarment from all Federal procurement and nonprocurement transactions (FAR Subpart 9.4 or 2 C.F.R. part 180); and (6) other administrative penalties including termination of SBIR/STTR awards.

Signature 	Date 02/29/2020/
Print Name John Peter Ramo	
Title CEO	
Business Name Boulder Learning Inc.	

G.6 North American Industry Classification System Code (NAICS) and DUNS Numbers

Certification G.6

This solicitation assumes North American Industry Classification System Code (NAICS) 541720 – Research and Development in the Social Sciences and Humanities. The offeror's SAM registration must include a self-certification that they are a Small Business under their chosen NAICS code. If the offeror is submitting their proposal under a different NAICS code, please provide it here:

Provide your DUNS number:
791197622

G.7 Additional Data

The next four (4) items are for statistical purposes only.

Certification G.7

Socially and Economically Disadvantaged SBC Certification –

Does the offeror qualify as a socially and economically disadvantaged SBC and meet the definition as stated in this program solicitation?

☐ Yes **X**No

Woman-owned SBC Certification –

Does the offeror qualify as a woman-owned SBC and meet the definition as stated in this program solicitation?

☐ Yes **X**No

HUBZone-owned SBC Certification –

Does the offeror qualify as a HUBZone-owned SBC and meet the definition as stated in this solicitation?

☐ Yes **X**No

The website listed below contains information about the SBA's HUBZone program:

<http://www.sba.gov/hubzone/>

Service Disabled Veteran Owned Small Business Certification –

Does the offeror qualify as a Service Disabled Veteran Owned Small Business?

☐ Yes **X**No

G.8 Manufacturing-Related Project Certification

This certification will serve as a tie-breaker.

Certification G.8

If R/R&D from an eventual Phase II award leads to a completed product or tool, is it possible that this product will be manufactured (e.g. production) on a wide scale basis?

☐ Yes **X**No

In cases where there is a tie in the award selection process, ED will give priority to projects that are manufacturing-related. This “tie-breaker” specification allows the ED program to apply an additional preference without compromising the quality standards or established criteria of the program.

G.9 Disclosure Permission Statement

Certification G.9

Will the offeror permit the Government to disclose the title and technical abstract page of the proposed project, plus the name, address, and telephone number of the corporate official of the offeror's firm, if the proposal does not result in an award, to concerns that may be interested in contacting you for further information?

XYes ☐No

G.10 Legend for Proprietary Information

Information contained in unsuccessful proposals will remain the property of the offeror. The Government may, however, retain copies of all proposals. Public release of information in any proposal submitted will be subject to existing statutory and regulatory requirements.

If proprietary information provided by an offeror in a proposal constitutes a trade secret, proprietary commercial or financial information, confidential personal information, or data affecting national security, it will be treated in confidence to the extent permitted by law. This information must be clearly marked by the offeror with the term "confidential proprietary information" and the following legend must appear in this section of the proposal:

Certification G.10

"These data shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed in whole or in part for any purpose other than evaluation of this proposal. If a contract is awarded to this offeror as a result of (or in connection with) the submission of these data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the contract and pursuant to applicable law. This restriction does not limit the Government's right to use information contained in the data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages (insert page number here) of this proposal."

XYes ☐No

The use of any other legend is unacceptable to the Government and may constitute grounds for removing the proposal from further consideration without assuming any liability for inadvertent

disclosure. The Government will limit dissemination of such information to/within official channels.

G.11 FAR 52.204-8 -- Annual Representations and Certifications. (Oct 2016)

(a)

(1) The North American Industry classification System (NAICS) code for this acquisition is **611710**.

(2) The small business size standard is **\$16.5M**.

(3) The small business size standard for a concern which submits an offer in its own name, other than on a construction or service contract, but which proposes to furnish a product which it did not itself manufacture, is 500 employees.

(b)

(1) If the provision at 52.204-7, System for Award Management, is included in this solicitation, paragraph (d) of this provision applies.

(2) If the provision at 52.204-7 is not included in this solicitation, and the offeror is currently registered in the System for Award Management (SAM), and has completed the Representations and Certifications section of SAM electronically, the offeror may choose to use paragraph (d) of this provision instead of completing the corresponding individual representations and certification in the solicitation. The offeror shall indicate which option applies by checking one of the following boxes:

☐ (i) Paragraph (d) applies.

☒ (ii) Paragraph (d) does not apply and the offeror has completed the individual representations and certifications in the solicitation.

(c)

(1) The following representations or certifications in SAM are applicable to this solicitation as indicated:

(i) 52.203-2, Certificate of Independent Price Determination. This provision applies to solicitations when a firm-fixed-price contract or fixed-price contract with economic price adjustment is contemplated, unless—

(A) The acquisition is to be made under the simplified acquisition procedures in Part 13;

(B) The solicitation is a request for technical proposals under two-step sealed bidding procedures; or

(C) The solicitation is for utility services for which rates are set by law or regulation.

(ii) 52.203-11, Certification and Disclosure Regarding Payments to Influence Certain Federal Transactions. This provision applies to solicitations expected to exceed \$150,000.

(iii) 52.204-3, Taxpayer Identification. This provision applies to solicitations that do not include the provision at 52.204-7, System for Award Management.

(iv) 52.204-5, Women-Owned Business (Other Than Small Business). This provision applies to solicitations that—

- (A) Are not set aside for small business concerns;
- (B) Exceed the simplified acquisition threshold; and
- (C) Are for contracts that will be performed in the United States or its outlying areas.

(v) 52.209-2, Prohibition on Contracting with Inverted Domestic Corporations—Representation.

(vi) 52.209-5; Certification Regarding Responsibility Matters. This provision applies to solicitations where the contract value is expected to exceed the simplified acquisition threshold.

(vii) 52.209-11, Representation by Corporations Regarding Delinquent Tax Liability or a Felony Conviction under any Federal Law. This provision applies to all solicitations.

(viii) 52.214-14, Place of Performance--Sealed Bidding. This provision applies to invitations for bids except those in which the place of performance is specified by the Government.

(ix) 52.215-6, Place of Performance. This provision applies to solicitations unless the place of performance is specified by the Government.

(x) 52.219-1, Small Business Program Representations (Basic & Alternate I). This provision applies to solicitations when the contract will be performed in the United States or its outlying areas.

- (A) The basic provision applies when the solicitations are issued by other than DoD, NASA, and the Coast Guard.

- (B) The provision with its Alternate I applies to solicitations issued by DoD, NASA, or the Coast Guard.

(xi) 52.219-2, Equal Low Bids. This provision applies to solicitations when contracting by sealed bidding and the contract will be performed in the United States or its outlying areas.

(xii) 52.222-22, Previous Contracts and Compliance Reports. This provision applies to solicitations that include the clause at 52.222-26, Equal Opportunity.

(xiii) 52.222-25, Affirmative Action Compliance. This provision applies to solicitations, other than those for construction, when the solicitation includes the clause at 52.222-26, Equal Opportunity.

(xiv) 52.222-38, Compliance with Veterans' Employment Reporting Requirements. This provision applies to solicitations when it is anticipated the contract award will exceed the simplified acquisition threshold and the contract is not for acquisition of commercial items.

(xv) 52.222-57, Representation Regarding Compliance with Labor Laws (Executive Order 13673). This provision applies to solicitations expected to exceed \$50 million which are issued from October 25, 2016 through April 24, 2020, and solicitations expected to exceed \$500,000, which are issued after April 24, 2020.

(xvi) 52.223-1, Biobased Product Certification. This provision applies to solicitations that require the delivery or specify the use of USDA-designated

items; or include the clause at 52.223-2, Affirmative Procurement of Biobased Products Under Service and Construction Contracts.

(xvii) 52.223-4, Recovered Material Certification. This provision applies to solicitations that are for, or specify the use of, EPA- designated items.

(xviii) 52.225-2, Buy American Certificate. This provision applies to solicitations containing the clause at 52.225-1.

(xix) 52.225-4, Buy American--Free Trade Agreements--Israeli Trade Act Certificate. (Basic, Alternates I, II, and III.) This provision applies to solicitations containing the clause at 52.225- 3.

(A) If the acquisition value is less than \$25,000, the basic provision applies.

(B) If the acquisition value is \$25,000 or more but is less than \$50,000, the provision with its Alternate I applies.

(C) If the acquisition value is \$50,000 or more but is less than \$77,533, the provision with its Alternate II applies.

(D) If the acquisition value is \$79,507 or more but is less than \$100,000, the provision with its Alternate III applies.

(xx) 52.225-6, Trade Agreements Certificate. This provision applies to solicitations containing the clause at 52.225-5.

(xxi) 52.225-20, Prohibition on Conducting Restricted Business Operations in Sudan--Certification. This provision applies to all solicitations.

(xxii) 52.225-25, Prohibition on Contracting with Entities Engaging in Certain Activities or Transactions Relating to Iran—Representation and Certification. This provision applies to all solicitations.

(xxiii) 52.226-2, Historically Black College or University and Minority Institution Representation. This provision applies to solicitations for research, studies, supplies, or services of the type normally acquired from higher educational institutions.

(2) The following representations or certifications are applicable as indicated by the Contracting Officer:

[Contracting Officer check as appropriate.]

[√] (i) 52.204-17, Ownership or Control of Offeror.

[√] (ii) 52.204-20, Predecessor of Offeror.

[N/A] (iii) 52.222-18, Certification Regarding Knowledge of Child Labor for Listed End Products.

[N/A] (iv) 52.222-48, Exemption from Application of the Service Contract Labor Standards to Contracts for Maintenance, Calibration, or Repair of Certain Equipment--Certification.

[N/A] (v) 52.222-52 Exemption from Application of the Service Contract Labor Standards to Contracts for Certain Services--Certification.

[N/A] (vi) 52.223-9, with its Alternate I, Estimate of Percentage of Recovered Material Content for EPA-Designated Products (Alternate I only).

[N/A] (vii) 52.227-6, Royalty Information.

- ___ (A) Basic.
 ___ (B) Alternate I.

[N/A] (viii) 52.227-15, Representation of Limited Rights Data and Restricted Computer Software.

(d) The offeror has completed the annual representations and certifications electronically via the SAM Web site accessed through <https://www.acquisition.gov>. After reviewing the SAM database information, the offeror verifies by submission of the offer that the representations and certifications currently posted electronically that apply to this solicitation as indicated in paragraph (c) of this provision have been entered or updated within the last 12 months, are current, accurate, complete, and applicable to this solicitation (including the business size standard applicable to the NAICS code referenced for this solicitation), as of the date of this offer and are incorporated in this offer by reference (see FAR 4.1201); except for the changes identified below *[offeror to insert changes, identifying change by clause number, title, date]*. These amended representation(s) and/or certification(s) are also incorporated in this offer and are current, accurate, and complete as of the date of this offer.

FAR Clause	Title	Date	Change

Any changes provided by the offeror are applicable to this solicitation only, and do not result in an update to the representations and certifications posted on SAM.

(End of Provision)

Certification G.11

The offeror affirms provision 52.204 above.

XYes ☐ No

G.12 FAR 52.222-54 – Employment Eligibility Verification (Oct 2015)

(a) *Definitions.* As used in this clause—

“Commercially available off-the-shelf (COTS) item”—

(1) Means any item of supply that is—

- (i) A commercial item (as defined in paragraph (1) of the definition at 2.101);
- (ii) Sold in substantial quantities in the commercial marketplace; and
- (iii) Offered to the Government, without modification, in the same form in which it is sold in the commercial marketplace; and

(2) Does not include bulk cargo, as defined in 46 U.S.C. 40102(4) such as agricultural products and petroleum products. Per 46 CFR 525.1(c)(2), “bulk cargo” means cargo that is loaded and carried in bulk onboard ship without mark or count, in a loose unpackaged form, having homogenous characteristics. Bulk cargo loaded into intermodal equipment, except LASH or Seabee barges, is subject to mark and count and, therefore, ceases to be bulk cargo.

“Employee assigned to the contract” means an employee who was hired after November 6, 1986 (after November 27, 2009, in the Commonwealth of the Northern Mariana Islands), who is directly performing work, in the United States, under a contract that is required to include the clause prescribed at 22.1803. An employee is not considered to be directly performing work under a contract if the employee—

- (1) Normally performs support work, such as indirect or overhead functions; and
- (2) Does not perform any substantial duties applicable to the contract.

“Subcontract” means any contract, as defined in 2.101, entered into by a subcontractor to furnish supplies or services for performance of a prime contract or a subcontract. It includes but is not limited to purchase orders, and changes and modifications to purchase orders.

“Subcontractor” means any supplier, distributor, vendor, or firm that furnishes supplies or services to or for a prime Contractor or another subcontractor.

“United States,” as defined in 8 U.S.C. 1101(a)(38), means the 50 States, the District of Columbia, Puerto Rico, Guam, the Commonwealth of the Northern Mariana Islands, and the U.S. Virgin Islands.

(b) Enrollment and verification requirements.

(1) If the Contractor is not enrolled as a Federal Contractor in E-Verify at time of contract award, the Contractor shall—

- (i) *Enroll.* Enroll as a Federal Contractor in the E-Verify program within 30 calendar days of contract award;
- (ii) *Verify all new employees.* Within 90 calendar days of enrollment in the E-Verify program, begin to use E-Verify to initiate verification of employment eligibility of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract, within 3 business days after the date of hire (but see paragraph (b)(3) of this section); and
- (iii) *Verify employees assigned to the contract.* For each employee assigned to the contract, initiate verification within 90 calendar days after date of enrollment or within 30 calendar days of the employee’s assignment to the contract, whichever date is later (but see paragraph (b)(4) of this section).

(2) If the Contractor is enrolled as a Federal Contractor in E-Verify at time of contract award, the Contractor shall use E-Verify to initiate verification of employment eligibility of—

(i) *All new employees.*

(A) *Enrolled 90 calendar days or more.* The Contractor shall initiate verification of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract within 3 business days after the date of hire (but see paragraph (b)(3) of this section); or

(B) *Enrolled less than 90 calendar days.* Within 90 calendar days after enrollment as a Federal Contractor in E-Verify, the Contractor shall initiate verification of all new hires of the Contractor, who are working in the United States, whether or not assigned to the contract, within 3 business days after the date of hire (but see paragraph (b)(3) of this section); or

(ii) *Employees assigned to the contract.* For each employee assigned to the contract, the Contractor shall initiate verification within 90 calendar days after date of contract award or within 30 days after assignment to the contract, whichever date is later (but see paragraph (b)(4) of this section).

(3) If the Contractor is an institution of higher education (as defined at 20 U.S.C. 1001(a)); a State or local government or the government of a Federally recognized Indian tribe; or a surety performing under a takeover agreement entered into with a Federal agency pursuant to a performance bond, the Contractor may choose to verify only employees assigned to the contract, whether existing employees or new hires. The Contractor shall follow the applicable verification requirements at (b)(1) or (b)(2), respectively, except that any requirement for verification of new employees applies only to new employees assigned to the contract.

(4) *Option to verify employment eligibility of all employees.* The Contractor may elect to verify all existing employees hired after November 6, 1986 (after November 27, 2009, in the Commonwealth of the Northern Mariana Islands), rather than just those employees assigned to the contract. The Contractor shall initiate verification for each existing employee working in the United States who was hired after November 6, 1986 (after November 27, 2009, in the Commonwealth of the Northern Mariana Islands), within 180 calendar days of—

(i) Enrollment in the E-Verify program; or

(ii) Notification to E-Verify Operations of the Contractor's decision to exercise this option, using the contact information provided in the E-Verify program Memorandum of Understanding (MOU).

(5) The Contractor shall comply, for the period of performance of this contract, with the requirement of the E-Verify program MOU.

(i) The Department of Homeland Security (DHS) or the Social Security Administration (SSA) may terminate the Contractor's MOU and deny access to the E-Verify system in accordance with the terms of the MOU. In such case, the Contractor will be referred to a suspension or debarment official.

(ii) During the period between termination of the MOU and a decision by the suspension or debarment official whether to suspend or debar, the Contractor is excused from its obligations under paragraph (b) of this clause. If the suspension

or debarment official determines not to suspend or debar the Contractor, then the Contractor must reenroll in E-Verify.

(c) *Web site.* Information on registration for and use of the E-Verify program can be obtained via the Internet at the Department of Homeland Security Web site: <http://www.dhs.gov/E-Verify> .

(d) *Individuals previously verified.* The Contractor is not required by this clause to perform additional employment verification using E-Verify for any employee—

- (1) Whose employment eligibility was previously verified by the Contractor through the E-Verify program;
- (2) Who has been granted and holds an active U.S. Government security clearance for access to confidential, secret, or top secret information in accordance with the National Industrial Security Program Operating Manual; or
- (3) Who has undergone a completed background investigation and been issued credentials pursuant to Homeland Security Presidential Directive (HSPD) -12, Policy for a Common Identification Standard for Federal Employees and Contractors.

(e) *Subcontracts.* The contractor shall include the requirements of this clause, including this paragraph (e) (appropriately modified for identification of the parties), in each subcontract that—

(1) *Is for—*

(i) Commercial or noncommercial services (except for commercial services that are part of the purchase of a COTS item (or an item that would be a COTS item, but for minor modifications), performed by the COTS provider, and are normally provided for that COTS item); or

(ii) Construction;

(2) Has a value of more than \$3,500; and

(3) Includes work performed in the United States.

(End of Clause)

Certification G.12

The offeror has read and understood the above clause in this section.

XYes ☐No

G.13 FAR 52.233-2 -- Service of Protest. (Sep 2006)

(a) Protests, as defined in section 33.101 of the Federal Acquisition Regulation, that are filed directly with an agency, and copies of any protests that are filed with the Government Accountability Office (GAO), shall be served on the Contracting Officer (addressed as follows) by obtaining written and dated acknowledgment of receipt from _Brandon Holding.

(b) The copy of any protest shall be received in the office designated above within one day of filing a protest with the GAO.

(End of Provision)

Certification G.13

The offeror has read and understood the above clause in this section.

X Yes ☐ No

G.14 FAR 52.233-3 -- Protest After Award. (Aug 1996)

(a) Upon receipt of a notice of protest (as defined in FAR 33.101) or a determination that a protest is likely (see FAR 33.102(d)), the Contracting Officer may, by written order to the Contractor, direct the Contractor to stop performance of the work called for by this contract. The order shall be specifically identified as a stop-work order issued under this clause. Upon receipt of the order, the Contractor shall immediately comply with its terms and take all reasonable steps to minimize the incurrence of costs allocable to the work covered by the order during the period of work stoppage. Upon receipt of the final decision in the protest, the Contracting Officer shall either --

(1) Cancel the stop-work order; or

(2) Terminate the work covered by the order as provided in the Default, or the Termination for Convenience of the Government, clause of this contract.

(b) If a stop-work order issued under this clause is canceled either before or after a final decision in the protest, the Contractor shall resume work. The Contracting Officer shall make an equitable adjustment in the delivery schedule or contract price, or both, and the contract shall be modified, in writing, accordingly, if --

(1) The stop-work order results in an increase in the time required for, or in the Contractor's cost properly allocable to, the performance of any part of this contract; and

(2) The Contractor asserts its right to an adjustment within 30 days after the end of the period of work stoppage; provided, that if the Contracting Officer decides the facts justify the action, the Contracting Officer may receive and act upon a proposal at any time before final payment under this contract.

(c) If a stop-work order is not canceled and the work covered by the order is terminated for the convenience of the Government, the Contracting Officer shall allow reasonable costs resulting from the stop-work order in arriving at the termination settlement.

(d) If a stop-work order is not canceled and the work covered by the order is terminated for default, the Contracting Officer shall allow, by equitable adjustment or otherwise, reasonable costs resulting from the stop-work order.

(e) The Government's rights to terminate this contract at any time are not affected by action taken under this clause.

(f) If, as the result of the Contractor's intentional or negligent misstatement, misrepresentation, or miscertification, a protest related to this contract is sustained, and the Government pays costs, as provided in FAR 33.102(b)(2) or 33.104(h)(1), the Government may require the Contractor to reimburse the Government the amount of such costs. In addition to any other remedy available, and pursuant to the requirements of Subpart 32.6, the Government may collect this debt by offsetting the amount against any payment due the Contractor under any contract between the Contractor and the Government.

(End of Clause)

Certification G.14

The offeror has read and understood the above clause in this section.

XYes ☐No

G. 15 FAR 52.233-4 -- Applicable Law for Breach of Contract Claim. (Oct 2004)

United States law will apply to resolve any claim of breach of this contract.

(End of clause)

Certification G.15

The offeror has read and understood the above clause in this section.

XYes ☐No

G.16 EDAR 3452.209-70 Conflict of Interest Certification (MAY 2011)

(a)(1) The contractor, subcontractor, employee, or consultant, by signing the form in this clause, certifies that, to the best of its knowledge and belief, there are no relevant facts or circumstances that could give rise to an organizational or personal conflict of interest, (*see* FAR Subpart 9.5 for organizational conflicts of interest) (or apparent conflict of interest), for the organization or any of its staff, and that the contractor, subcontractor, employee, or consultant has disclosed all such relevant information if such a conflict of interest appears to exist to a reasonable person with knowledge of the relevant facts (or if such a person would question the impartiality of the contractor, subcontractor, employee, or consultant). Conflicts may arise in the following situations:

(i) *Unequal access to information.* A potential contractor, subcontractor, employee, or consultant has access to non-public information through its performance on a government contract.

(ii) *Biased ground rules.* A potential contractor, subcontractor, employee, or consultant has worked, in one government contract, or program, on the basic structure or ground rules of another government contract.

(iii) *Impaired objectivity.* A potential contractor, subcontractor, employee, or consultant, or member of their immediate family (spouse, parent, or child) has financial or other interests that would impair, or give the appearance of impairing, impartial judgment in the evaluation of government programs, in offering advice or recommendations to the government, or in providing technical assistance or other services to recipients of Federal funds as part of its contractual responsibility. "Impaired objectivity" includes but is not limited to the following situations that would cause a reasonable person with knowledge of the relevant facts to question a person's objectivity:

(A) Financial interests or reasonably foreseeable financial interests in or in connection with products, property, or services that may be purchased by an educational agency, a person, organization, or institution in the course of implementing any program administered by the Department;

(B) Significant connections to teaching methodologies or approaches that might require or encourage the use of specific products, property, or services; or

(C) Significant identification with pedagogical or philosophical viewpoints that might require or encourage the use of a specific curriculum, specific products, property, or services.

(2) Offerors must provide the disclosure described above on any actual or potential conflict of interest (or apparent conflict of interest) regardless of their opinion that such a conflict or potential conflict (or apparent conflict of interest) would not impair their objectivity.

(3) In a case in which an actual or potential conflict (or apparent conflict of interest) is disclosed, the Department will take appropriate actions to eliminate or address the actual or potential conflict, including but not limited to mitigating or neutralizing the conflict, when appropriate, through such means as ensuring a balance of views, disclosure with the appropriate disclaimers, or by restricting or

modifying the work to be performed to avoid or reduce the conflict. In this clause, the term “potential conflict” means reasonably foreseeable conflict of interest.

(b) The contractor, subcontractor, employee, or consultant agrees that if “impaired objectivity”, or an actual or potential conflict of interest (or apparent conflict of interest) is discovered after the award is made, it will make a full disclosure in writing to the contracting officer. This disclosure shall include a description of actions that the contractor has taken or proposes to take to avoid, mitigate, or neutralize the actual or potential conflict (or apparent conflict of interest).

(c) *Remedies.* The Government may terminate this contract for convenience, in whole or in part, if it deems such termination necessary to avoid the appearance of a conflict of interest. If the contractor was aware of a potential conflict of interest prior to award or discovered an actual or potential conflict after award and did not disclose or misrepresented relevant information to the contracting officer, the Government may terminate the contract for default, or pursue such other remedies as may be permitted by law or this contract. These remedies include imprisonment for up to five years for violation of 18 U.S.C. 1001 and fines of up to \$5000 for violation of 31 U.S.C. 3802. Further remedies include suspension or debarment from contracting with the Federal government. The contractor may also be required to reimburse the Department for costs the Department incurs arising from activities related to conflicts of interest. An example of such costs would be those incurred in processing Freedom of Information Act requests related to a conflict of interest.

(d) In cases where remedies short of termination have been applied, the contractor, subcontractor, employee, or consultant agrees to eliminate the organizational conflict of interest, or mitigate it to the satisfaction of the contracting officer.

(e) The contractor further agrees to insert in any subcontract or consultant agreement hereunder, provisions that conform substantially to the language of this clause, including specific mention of potential remedies and this paragraph (e).

(f) *Conflict of Interest Certification.*

The offeror, [insert name of offeror], hereby certifies that, to the best of its knowledge and belief, there are no present or currently planned interests (financial, contractual, organizational, or otherwise) relating to the work to be performed under the contract or task order resulting from Request for Proposal No. [insert number] that would create any actual or potential conflict of interest (or apparent conflicts of interest) (including conflicts of interest for immediate family members: spouses, parents, children) that would impinge on its ability to render impartial, technically sound, and objective assistance or advice or result in it being given an unfair competitive advantage. In this clause, the term “potential conflict” means reasonably foreseeable conflict of interest.

The offeror further certifies that it has and will continue to exercise due diligence in identifying and removing or mitigating, to the Government's satisfaction, such conflict of interest (or apparent conflict of interest).

Offeror's Name Boulder Learning, Inc.

RFP/Contract No. 91990020R0006

Signature 

Title Principle Investigator/Methodologist

Date 2/29/20

(End of Provision)

G.17 EDAR 3452.209-71 Conflict of Interest (MAY 2011)

(a)(1) The contractor, subcontractor, employee, or consultant, has certified that, to the best of its knowledge and belief, there are no relevant facts or circumstances that could give rise to an organizational or personal conflict of interest (*see* FAR Subpart 9.5 for organizational conflicts of interest) (or apparent conflict of interest) for the organization or any of its staff, and that the contractor, subcontractor, employee, or consultant has disclosed all such relevant information if such a conflict of interest appears to exist to a reasonable person with knowledge of the relevant facts (or if such a person would question the impartiality of the contractor, subcontractor, employee, or consultant). Conflicts may arise in the following situations:

(i) *Unequal access to information* —A potential contractor, subcontractor, employee, or consultant has access to non-public information through its performance on a government contract.

(ii) *Biased ground rules* —A potential contractor, subcontractor, employee, or consultant has worked, in one government contract, or program, on the basic structure or ground rules of another government contract.

(iii) *Impaired objectivity* —A potential contractor, subcontractor, employee, or consultant, or member of their immediate family (spouse, parent, or child) has financial or other interests that would impair, or give the appearance of impairing, impartial judgment in the evaluation of government programs, in offering advice or recommendations to the government, or in providing technical assistance or other services to recipients of Federal funds as part of its contractual responsibility.

“Impaired objectivity” includes but is not limited to the following situations that would cause a reasonable person with knowledge of the relevant facts to question a person's objectivity:

(A) Financial interests or reasonably foreseeable financial interests in or in connection with products, property, or services that may be purchased by an educational agency, a person, organization, or institution in the course of implementing any program administered by the Department;

(B) Significant connections to teaching methodologies that might require or encourage the use of specific products, property, or services; or

(C) Significant identification with pedagogical or philosophical viewpoints that might require or encourage the use of a specific curriculum, specific products, property, or services.

(2) Offerors must provide the disclosure described above on any actual or potential conflict (or apparent conflict of interest) of interest regardless of their opinion that such a conflict or potential conflict (or apparent conflict of interest) would not impair their objectivity.

(3) In a case in which an actual or potential conflict (or apparent conflict of interest) is disclosed, the Department will take appropriate actions to eliminate or address the actual or potential conflict (or apparent conflict of interest), including but not limited to mitigating or neutralizing the conflict, when appropriate, through such means as ensuring a balance of views, disclosure with the appropriate disclaimers, or by restricting or modifying the work to be performed to avoid or reduce the conflict. In this clause, the term “potential conflict” means reasonably foreseeable conflict of interest.

(b) The contractor, subcontractor, employee, or consultant agrees that if “impaired objectivity”, or an actual or potential conflict of interest (or apparent conflict of interest) is discovered after the award is made, it will make a full disclosure in writing to the contracting officer. This disclosure shall include a description of actions that the contractor has taken or proposes to take, after consultation with the contracting officer, to avoid, mitigate, or neutralize the actual or potential conflict (or apparent conflict of interest).

(c) *Remedies.* The Government may terminate this contract for convenience, in whole or in part, if it deems such termination necessary to avoid the appearance of a conflict of interest. If the contractor was aware of a potential conflict of interest prior to award or

discovered an actual or potential conflict (or apparent conflict of interest) after award and did not disclose or misrepresented relevant information to the contracting officer, the Government may terminate the contract for default, or pursue such other remedies as may be permitted by law or this contract. These remedies include imprisonment for up to five years for violation of 18 U.S.C. 1001 and fines of up to \$5,000 for violation of 31 U.S.C. 3802. Further remedies include suspension or debarment from contracting with the Federal government. The contractor may also be required to reimburse the Department for costs the Department incurs arising from activities related to conflicts of interest. An example of such costs would be those incurred in processing Freedom of Information Act requests related to a conflict of interest.

(d) In cases where remedies short of termination have been applied, the contractor, subcontractor, employee, or consultant agrees to eliminate the organizational conflict of interest, or mitigate it to the satisfaction of the contracting officer.

(e) The contractor further agrees to insert in any subcontract or consultant agreement hereunder, provisions that conform substantially to the language of this clause, including specific mention of potential remedies and this paragraph (e).

(End of Clause)

The offeror has read and understood the above clause in this section.

XYes ☐No

G.18 EDAR 3452.239–72 DEPARTMENT SECURITY REQUIREMENTS (DEVIATION)

a) The contractor and its subcontractors shall comply with Department security policy requirements as set forth in the "Bidder's Security Package: Security Requirements for Contractors Doing Business with the Department of Education" at www.ed.gov/fund/contract/about/bsp.html.

(b) Contractor employees who will have access to proprietary or sensitive ED information including "Controlled Unclassified Information" as defined in 32 CFR Part 2002, or ED IT systems, systems maintained on behalf of ED, ED facilities or space, and/or perform duties in a school or in a location where children are present, must undergo a personnel security screening and a favorable determination and are subject to reinvestigation as described in the "Contractor

Vetting Security Requirements.” Compliance with these “Contractor Vetting Security Requirements,” as amended, is required.

(c) The type of security investigation required to commence work on an ED contract is dictated by the position designation determination assigned by ED. All ED contractor positions are designated commensurate with their position risk/sensitivity, in accordance with Title 5 Code of Federal Regulations (5 CFR 731.106) and OPM’s Position Designation Tool (PDT) located at: <https://pdt.nbis.mil/>. The position designation determines the risk level and the corresponding level of background investigations required.

(d) The contractor shall comply with all contractor position designations established by ED.

(e) The following are the contractor employee positions required under this contract and their designated risk levels:

High Risk (HR): *(Not Applicable)*

Moderate Risk (MR): *X(Principal Investigator, Education Researcher)*

Low Risk (LR): *(Not Applicable)*

NOTES:

- The High Risk (HR) designation is *not applicable* for the ED/IES SBIR program.
- The Moderate Risk (MR) designation refers to individuals on the contract who are required to be cleared at Moderate risk if they work on the project 30 days or more, visit a school or come in contact with students, and have access to any personally identifiable information (PII) about the students.
- The Low Risk (LR) designation refers to individuals on the contract who are required to be cleared as Low risk if they work on the project 30 days or more and visit a school or come in contact with students, but do not have access to any personally identifiable information (PII) about the students.
- If individuals who perform work on the contract do not work for a total of 30 days, do not go to a school, come in contact with a student, or do not have access to PII, then NO clearance is needed.

Position Designation Records (PDRs) for ED/IES SBIR Contracts

Personnel Position	Description of Relevant Responsibilities	Risk Level
Principal Investigator	Leads the managerial team in planning, conducting, monitoring, and evaluating the work. Has access to sensitive but unclassified or Privacy Act-protected information.	MR
Education Researcher (Analyst, Programmer, Statistician, Research Assistant)	Prepares, programs, and/or analyzes data that include sensitive but unclassified or Privacy Act-protected information.	MR
Data Collector	Visits schools and administers an electronic survey to students (but does not have access to student data).	MR

(f) For performance-based contracts where ED has not identified required labor categories for contractor positions, ED considers the risk sensitivity of the services to be performed and the access to ED facilities and systems that will be required during performance, to determine the uniform contractor position risk level designation for all contractor employees who will be providing services under the contract. The uniform contractor position risk level designation applicable to this performance-based contract is:

(Moderate)

(g) Only U.S. citizens will be eligible for employment on contracts requiring a Moderate Risk/Public Trust, High Risk/Public Trust, or a National Security designation.

(h) Permanent resident aliens may be eligible for employment on contracts requiring Low Risk/Public Trust access.

(i) An approved waiver, in accordance with “Contractor Vetting Security Requirements” is required for any exception to the requirements of paragraphs (h) and (i) above.

(j) The Contractor shall-

- (1) Comply with the Principal Office (PO) processing requirements for personnel security screening,
- (2) Ensure that no contractor employee is placed in a higher risk position than for which he or she is approved,
- (3) Ensure contractor employees submit required security forms for reinvestigation in accordance with the timeframes set forth in the “Contractor Vetting Security Requirements,”
- (4) Report to the COR any information (i.e., personal conduct, criminal conduct, financial difficulties, etc.) that would raise a concern about the suitability of a contractor or whether a contractor employee’s continued employment would promote the efficiency of the service or violate the public trust,
- (5) Protect sensitive and Privacy Act-protected, including “Controlled Unclassified Information” as defined in 32 CFR Part 2002, from unauthorized access, use or misuse by its contractor employees, prevent unauthorized access by others, and report any instances of unauthorized access, use or misuse to the COR,
- (6) Report to the COR within two business days any removal of a contractor employee from a contract; or within one business day if removed for cause,
- (7) Return a PIV ID to the COR within seven business days of the contractor employee’s departure,
- (8) Report to COR of any job activities that contractor employee has brought to their attention that they believe could result in a change in the contractor employee’s position or the need for increased security access; and
- (9) Ensure that any ED information processed, stored, or transmitted by the contractor will not be used or redistributed except as specified in the contract.

(k) Performance of this contract will ☒ will not ☐ involve access to ED IT systems and/or systems maintained on behalf of ED. For contracts that require access to ED IT systems and/or systems maintained on behalf of ED, the Information Security Categorization applicable to each security objective has been determined to be:

Confidentiality: ☐ Low ☒ Moderate ☐ High (Contracting Officer to complete)

Integrity: ☐ Low ☒ Moderate ☐ High (Contracting Officer to complete)

Availability: ☐ Low ☒ Moderate ☐ High (Contracting Officer to complete)

Overall Risk Level: ☐ Low ☒ Moderate ☐ High (Contracting Officer to complete)

(l) Performance of this contract ☒ does involve ☐ does not involve Personally Identifiable information (PII) or “Controlled Unclassified Information” as defined in 32 CFR Part 2002. The Confidentiality Impact Level of such information has been determined to be:

☐ Not Applicable ☐ Low ☒ Moderate ☐ High.

(m) Failure to comply with any of the personnel, privacy, and cyber security requirements may result in a termination of the contract for default or cause.

Contractor Personnel Listing Form

Employee's Last Name	Employee's First Name	Employee's Middle Name (or Initial)	Contractor's Company Name	Sub-contractor's Company Name (if applicable)	Contract Name	Work Address	Employee Position	Clearance Level	Security Clearance Status	Date Assigned to Clearance Position	CORs Name

Security Clearance Application Initiation Form

Social Security Number (insert dashes, e.g., 123-45-6789)	Employee's Last Name	Employee's First Name	Employee's Middle Name (or Initial)	Suffix (e.g., MD) if there is no suffix leave blank, do not insert NA	Date of Birth (e.g., 1/1/1920)	City of Birth (e.g., Raleigh)	State of Birth (e.g., NC) if foreign leave blank, do not insert NA	Country of Birth (e.g., United States, Brazil) do not abbreviate	Email address (e.g., 123smith@att.net)	Phone Number (insert dashes, e.g., 501-555-1212, do not use periods)

(End of Clause)

(End of Clause)

Certification G.18

The offeror has read and understood the above clause in this section.

XYes ☐No

G.19 FAR 52.204-10 – Reporting Executive Compensation and First-Tier Subcontract Awards (Oct 2016)

(a) *Definitions.* As used in this clause:

“Executive” means officers, managing partners, or any other employees in management positions.

“First-tier subcontract” means a subcontract awarded directly by the Contractor for the purpose of acquiring supplies or services (including construction) for performance of a prime contract. It does not include the Contractor’s supplier agreements with vendors, such as long-term arrangements for materials or supplies that would benefit multiple contracts and/or the costs of which are normally applied to a Contractor’s general and administrative expenses or indirect cost.

“Month of award” means the month in which a contract is signed by the Contracting Officer or the month in which a first-tier subcontract is signed by the Contractor.

“Total compensation” means the cash and noncash dollar value earned by the executive during the Contractor’s preceding fiscal year and includes the following (for more information see 17 CFR 229.402(c)(2)):

- (1) Salary and bonus.
- (2) Awards of stock, stock options, and stock appreciation rights. Use the dollar amount recognized for financial statement reporting purposes with respect to the fiscal year in accordance with the Financial Accounting Standards Board’s Accounting Standards Codification (FASB ASC) 718, Compensation-Stock Compensation.
- (3) Earnings for services under non-equity incentive plans. This does not include group life, health, hospitalization or medical reimbursement plans that do not discriminate in favor of executives, and are available generally to all salaried employees.
- (4) Change in pension value. This is the change in present value of defined benefit and actuarial pension plans.
- (5) Above-market earnings on deferred compensation which is not tax-qualified.
- (6) Other compensation, if the aggregate value of all such other compensation (*e.g.*, severance, termination payments, value of life insurance paid on behalf of the employee, perquisites or property) for the executive exceeds \$10,000.

(b) Section 2(d)(2) of the Federal Funding Accountability and Transparency Act of 2006 (Pub. L. No. 109-282), as amended by section 6202 of the Government Funding Transparency Act of 2008 (Pub. L. 110-252), requires the Contractor to report information on subcontract awards. The law requires all reported information be made public, therefore, the Contractor is responsible for notifying its subcontractors that the required information will be made public.

(c) Nothing in this clause required the disclosure of classified information.

(d)

(1) Executive compensation of the prime contractor. As a part of its annual registration requirement in the System for Award Management (SAM) database (FAR provision 52.204-7), the Contractor shall report the names and total compensation of each of the five most highly compensated executives for its preceding completed fiscal year, if—

(i) In the Contractor's preceding fiscal year, the Contractor received—

(A) 80 percent or more of its annual gross revenues from Federal contracts (and subcontracts), loans, grants (and subgrants), cooperative agreements, and other forms of Federal financial assistance; and

(B) \$25,000,000 or more in annual gross revenues from Federal contracts (and subcontracts), loans, grants (and subgrants), cooperative agreements, and other forms of Federal financial assistance; and

(ii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/execomp.htm>.)

(2) First-tier subcontract information. Unless otherwise directed by the contracting officer, or as provided in paragraph (g) of this clause, by the end of the month following the month of award of a first-tier subcontract with a value of \$30,000 or more, the Contractor shall report the following information at <http://www.fsr.gov> for that first tier subcontract. (The Contractor shall follow the instruction at <http://www.fsr.gov> to report the data.)

(i) Unique entity identifier for the subcontractor receiving the award and for the subcontractor's parent company, if the subcontractor has a parent company.

(ii) Name of the subcontractor.

(iii) Amount of the subcontract award.

(iv) Date of the subcontract award.

(v) A description of the products or services (including construction) being provided under the subcontract, including the overall purpose and expected outcomes or results of the subcontract.

(vi) Subcontract number (the subcontract number assigned by the Contractor).

(vii) Subcontractor's physical address including street address, city, state, and country. Also include the nine-digit zip code and congressional district.

(viii) Subcontractor's primary performance location including street address, city, state, and country. Also include the nine-digit zip code and congressional district.

(ix) The prime contract number, and order number if applicable.

(x) Awarding agency name and code.

(xi) Funding agency name and code.

(xii) Government contracting office code.

(xiii) Treasury account symbol (TAS) as reported in FPDS.

(xiv) The applicable North American Industry Classification System code (NAICS).

(3) Executive compensation of the first-tier subcontractor. Unless otherwise directed by the Contracting Officer, by the end of the month following the month of award of a first-tier subcontract with a value of \$30,000 or more, and annually thereafter (calculated from the prime contract award date), the Contractor shall report the names and total compensation of each of the five most highly compensated executives for that first-tier subcontractor for the first-tier subcontractor's preceding completed fiscal year at <https://www.fsrs.gov> , if—

(i) In the subcontractor's preceding fiscal year, the subcontractor received—

(A) 80 percent or more of its annual gross revenues from Federal contracts (and subcontracts), loans, grants (and subgrants), cooperative agreements, and other forms of Federal financial assistance; and

(B) \$25,000,000 or more in annual gross revenues from Federal contracts (and subcontracts), loans, grants (and subgrants), cooperative agreements and other forms of Federal financial assistance; and

(ii) The public does not have access to information about the compensation of the executives through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986. (To determine if the public has access to the compensation information, see the U.S. Security and Exchange Commission total compensation filings at <http://www.sec.gov/answers/execomp.htm>.)

(e) The Contractor shall not split or break down first-tier subcontract awards to a value less than \$30,000 to avoid the reporting requirements in paragraph (d) of this clause.

(f) The Contractor is required to report information on a first-tier subcontract covered by paragraph (d) when the subcontract is awarded. Continued reporting on the same subcontract is not required unless one of the reported data elements changes during the performance of the subcontract. The Contractor is not required to make further reports after the first-tier subcontract expires.

(g)

(1) If the Contractor in the previous tax year had gross income, from all sources, under \$300,000, the Contractor is exempt from the requirement to report subcontractor awards.

(2) If a subcontractor in the previous tax year had gross income from all sources under \$300,000, the Contractor does not need to report awards for that subcontractor

(h) The FSRS database at <http://www.fsrs.gov> will be prepopulated with some information from SAM and FPDS databases. If FPDS information is incorrect, the contractor should notify the contracting officer. If the SAM database information is incorrect, the contractor is responsible for correcting this information.

(End of clause)

Certification G.19

The offeror has read and understood the above clause in this section.

XYes ☐No

G.20 FAR 52.209-5 Certification Regarding Responsibility Matters (Oct 2015)

(a)

(1) The Offeror certifies, to the best of its knowledge and belief, that --

(i) The Offeror and/or any of its Principals --

(A) Are not presently debarred, suspended, proposed for debarment, or declared ineligible for the award of contracts by any Federal agency;

(B) Have not within a three-year period preceding this offer, been convicted of or had a civil judgment rendered against them for: commission of fraud or a criminal offense in connection with obtaining, attempting to obtain, or performing a public (Federal, State, or local) contract or subcontract; violation of Federal or State antitrust statutes relating to the submission of offers; or commission of embezzlement, theft, forgery, bribery, falsification or destruction of records, making false statements, tax evasion, violating Federal criminal tax laws, or receiving stolen property (if offeror checks "have", the offeror shall also see 52.209-7, if included in this solicitation); and

(C) Are not presently indicted for, or otherwise criminally or civilly charged by a governmental entity with, commission of any of the offenses enumerated in paragraph (a)(1)(i)(B) of this provision; and

(D) Have not, within a three-year period preceding this offer, been notified of any delinquent Federal taxes in an amount that exceeds \$3,500 for which the liability remains unsatisfied.

(1) Federal taxes are considered delinquent if both of the following criteria apply:

(i) *The tax liability is finally determined.* The liability is finally determined if it has been assessed. A liability is not finally determined if there is a

pending administrative or judicial challenge. In the case of a judicial challenge to the liability, the liability is not finally determined until all judicial appeal rights have been exhausted.

(ii) The taxpayer is delinquent in making payment.

A taxpayer is delinquent if the taxpayer has failed to pay the tax liability when full payment was due and required. A taxpayer is not delinquent in cases where enforced collection action is precluded.

(2) Examples.

(i) The taxpayer has received a statutory notice of deficiency, under I.R.C. §6212, which entitles the taxpayer to seek Tax Court review of a proposed tax deficiency. This is not a delinquent tax because it is not a final tax liability. Should the taxpayer seek Tax Court review, this will not be a final tax liability until the taxpayer has exercised all judicial appeal rights.

(ii) The IRS has filed a notice of Federal tax lien with respect to an assessed tax liability, and the taxpayer has been issued a notice under I.R.C. §6320 entitling the taxpayer to request a hearing with the IRS Office of Appeals contesting the lien filing, and to further appeal to the Tax Court if the IRS determines to sustain the lien filing. In the course of the hearing, the taxpayer is entitled to contest the underlying tax liability because the taxpayer has had no prior opportunity to contest the liability. This is not a delinquent tax because it is not a final tax liability. Should the taxpayer seek tax court review, this will not be a final tax liability until the taxpayer has exercised all judicial appeal rights.

(iii) The taxpayer has entered into an installment agreement pursuant to I.R.C. §6159. The taxpayer is making timely payments and is in full compliance with the agreement terms. The taxpayer is not delinquent because the taxpayer is not currently required to make full payment.

(iv) The taxpayer has filed for bankruptcy protection. The taxpayer is not delinquent because enforced collection action is stayed under 11 U.S.C. 362 (the Bankruptcy Code).

(ii) The Offeror has not, within a three-year period preceding this offer, had one or more contracts terminated for default by any Federal agency.

(2) "Principal," for the purposes of this certification, means an officer; director; owner; partner; or a person having primary management or supervisory responsibilities within a business entity (e.g., general manager; plant manager; head of a division or business segment; and similar positions).

This Certification Concerns a Matter Within the Jurisdiction of an Agency of the United States and the Making of a False, Fictitious, or Fraudulent Certification May Render the Maker Subject to Prosecution Under Section 1001, Title 18, United States Code.

(b) The Offeror shall provide immediate written notice to the Contracting Officer if, at any time prior to contract award, the Offeror learns that its certification was erroneous when submitted or has become erroneous by reason of changed circumstances.

(c) A certification that any of the items in paragraph (a) of this provision exists will not necessarily result in withholding of an award under this solicitation. However, the certification will be considered in connection with a determination of the Offeror's responsibility. Failure of the Offeror to furnish a certification or provide such additional information as requested by the Contracting Officer may render the Offeror non responsible.

(d) Nothing contained in the foregoing shall be construed to require establishment of a system of records in order to render, in good faith, the certification required by paragraph (a) of this provision. The knowledge and information of an Offeror is not required to exceed that which is normally possessed by a prudent person in the ordinary course of business dealings.

(e) The certification in paragraph (a) of this provision is a material representation of fact upon which reliance was placed when making award. If it is later determined that the Offeror knowingly rendered an erroneous certification, in addition to other remedies available to the Government, the Contracting Officer may terminate the contract resulting from this solicitation for default.

(End of Provision)

The offeror has read and understood the above provision in this section.

X Yes ☐ No

G.21 FAR 52.209-7 – Information Regarding Responsibility Matters (Jul 2013)

(a) *Definitions.* As used in this provision—

“Administrative proceeding” means a non-judicial process that is adjudicatory in nature in order to make a determination of fault or liability (*e.g.*, Securities and Exchange Commission Administrative Proceedings, Civilian Board of Contract Appeals Proceedings, and Armed Services Board of Contract Appeals Proceedings). This includes administrative proceeding at the Federal and State level but only in connection with performance of a Federal contract or grant. It does not include agency actions such as contract audits, site visits, corrective plans, or inspection of deliverables.

“Federal contracts and grants with total value greater than \$10,000,000” means—

(1) The total value of all current, active contracts and grants, including all priced options; and

(2) The total value of all current, active orders including all priced options under indefinite-delivery, indefinite-quantity, 8(a), or requirements contracts (including task and delivery and multiple-award Schedules).

“Principal” means an officer, director, owner, partner, or a person having primary management or supervisory responsibilities within a business entity (*e.g.*, general manager; plant manager; head of a division or business segment; and similar positions).

(b) The offeror does not have current active Federal contracts and grants with total value greater than \$10,000,000.

(c) If the offeror checked “has” in paragraph (b) of this provision, the offeror represents, by submission of this offer, that the information it has entered in the Federal Awardee Performance and Integrity Information System (FAPIS) is current, accurate, and complete as of the date of submission of this offer with regard to the following information:

(1) Whether the offeror, and/or any of its principals, has or has not, within the last five years, in connection with the award to or performance by the offeror of a Federal contract or grant, been the subject of a proceeding, at the Federal or State level that resulted in any of the following dispositions:

(i) In a criminal proceeding, a conviction.

(ii) In a civil proceeding, a finding of fault and liability that results in the payment of a monetary fine, penalty, reimbursement, restitution, or damages of \$5,000 or more.

(iii) In an administrative proceeding, a finding of fault and liability that results in—

(A) The payment of a monetary fine or penalty of \$5,000 or more; or

(B) The payment of a reimbursement, restitution, or damages in excess of \$100,000.

(iv) In a criminal, civil, or administrative proceeding, a disposition of the matter by consent or compromise with an acknowledgment of fault by the Contractor if the proceeding could have led to any of the outcomes specified in paragraphs (c)(1)(i), (c)(1)(ii), or (c)(1)(iii) of this provision.

(2) If the offeror has been involved in the last five years in any of the occurrences listed in (c)(1) of this provision, whether the offeror has provided the requested information with regard to each occurrence.

(d) The offeror shall post the information in paragraphs (c)(1)(i) through (c)(1)(iv) of this provision in FAPIIS as required through maintaining an active registration in the System for Award Management database via <https://www.acquisition.gov> (see 52.204-7).

(End of provision)

Certification G.21

The offeror has read and understood the above provision in this section.

XYes ☐ No

G.22 306-1b Invoice and Contract Financing Requests Submission – Invoice Processing Platform (IPP) (May 2018)

(a) Payments shall be rendered in accordance with the identified payment schedule(s), and any other incorporated payment clause(s), specified rates, and/or fixed price amounts.

(b)The contractor shall submit invoices electronically by means of the web-based system, Invoice Processing Platform (IPP) that can be accessed at: <http://www.ipp.gov/>. All submitted invoices must be accompanied by supporting documentation in accordance with the contract's terms and conditions. The supporting documentation shall be submitted in the following formats: Adobe Acrobat (pdf), Microsoft Word (doc), Pictures (jpeg), Microsoft Excel (excel), and Microsoft Outlook message (msg).

(c)Invoice Number Format – The format for the invoice shall be the contract number followed by the invoice number. The invoice number shall have no spaces, dashes, or other special characters. The invoice number cannot exceed 21 characters. Examples of acceptable invoice number formats are as follows:

Contract Number: 91990018C0001

Contractor's Invoice No.: 15897126341

IPP Invoice No.: 91990018C000115897126 (Note that the last three digits ["341"] from the Contractor's invoice number were removed due to the 21 character limitation.)

Contractor's invoice number were removed due to the 21 character limitation.)

(d)If the Contractor has not already established an IPP account that is active, an IPP account will be automatically created on behalf of the Contractor. The automatically created IPP account will be issued to the Designated Primary Administrator, which will be the individual that has been identified in the "Accounts Receivable POC" Section of the Contractor's System for Award Management (SAM) registration located at <https://www.sam.gov/portal/public/SAM/>.

(e)Within ten (10) business days of the Contractor entering or updating the Accounts Receivable POC information within the Contractor's SAM registration, the Designated Primary Administrator will receive an email from the IPP Customer Support Team containing the Designated Primary Administrator's IPP username. Within 24 hours of receiving the initial email, the Designated Primary Administrator will receive a second email containing their IPP password. Once both emails have been received, the Designated Primary Administrator must log into IPP and complete the registration process.

(f)The Contractor's Designated Primary Administrator will be authorized to further designate other administrators under the Contractor's IPP account who may submit invoices on behalf of the Contractor.

(g)The Contractor must ensure that the "Accounts Receivable POC" Section of its SAM registration is accurate and up to date.

(h) In the event that an invoice is rejected, the contractor shall make the necessary corrections and resubmit the invoice by means of IPP. Any questions, concerns, or issues regarding the use of IPP should be directed to IPP Customer Support Team, as identified at <http://www.ipp.gov/>
(End of clause)

Certification G.22

The offeror has read and understood the above clause under this section.

XYes ☐No

G.23 Clause 31.205.70 - Food Costs

No food may be provided under this contract or in association with this contract unless consent is provided below. The cost of food under this contract is unallowable unless the contractor receives written consent from the Contracting Officer prior to the incurrence of the cost. If the contractor wishes to be reimbursed for a food cost, it must make a request in writing at least 21 days prior to the day that costs would be incurred. The contractor shall include in its request the following: the purpose of the event at which the food will be served, why the food is integral to fulfill a government requirement in the contract, and the proposed costs. The lack of a timely response from the Contracting Officer shall not constitute constructive acceptance of the allowability of the proposed charge. Consent is hereby given to the contractor to: Funds are not authorized for the use of food.

Meetings – Include, but are not limited to, non-routine formal conferences, congresses, conventions, seminars, symposiums, workshops, with a published and substantive agenda with scheduled speakers and concern matters of topical interest to participants from multiple agencies and/or non-governmental participants. As used in this directive, “meetings” do not include purely internal routine business meetings primarily involving discussions of day-to-day internal procedures or operations such as, but not limited to, ED-sponsored staff meetings, all-hands meetings, working groups, retreats, etc.

Certification G.24

The offeror has read and understood the above clause in this section.

XYes ☐No

G.25 Clause 31.205.71 - Travel Costs

No invitational travel (defined as: Official government travel conducted by a non-federal employee in order to provide a “direct service” [i.e. presenting on a topic, serving as a facilitator, serving on a Federal Advisory Committee Act, or advising in an area of expertise] to the government) may be provided under this contract or in association with this contract unless consent is provided below. The cost of invitational travel under this contract is unallowable unless the contractor receives written consent from the Contracting Officer prior to the

incurrence of the cost. If the contractor wishes to be reimbursed for a cost related to invitational travel, it must make a request in writing at least 21 days prior to the day that costs would be incurred. The contractor shall include in its request the following: why the invitational travel cost is integral to fulfill a government requirement in the contract, and the proposed cost that must be in accordance with federal travel regulations. The lack of a timely response from the Contracting Officer shall not constitute constructive acceptance of the allow ability of the proposed charge. Consent is hereby given to the contractor to: Funds are not authorized for the use of travel.

Certification G.25

The offeror has read and understood the above clause in this section.

XYes ☐No

G.26 Clause 313.237-72 Clearance of Conferences/Meetings (DECEMBER 2012)

Any hotel/venue contract that the Contractor negotiates must be reviewed by and receive concurrence from an Event Services Team member prior to final agreement. The Event Services staff can be contacted at (202) 401-3679 or event.services@ed.gov. Comps: The Contractor does not have authority to negotiate or accept room upgrades for Department or Contractor staff. However, the Contractor is authorized to exercise its best efforts to obtain other Comps of necessary items/services that the Department would otherwise seek to procure in furtherance of the conference/meeting (i.e., meeting rooms, sleeping rooms, audio-visual equipment, etc.) Dual Compensation: Contractors are prohibited from receiving compensation from both the Department and any other source for conference planning performed pursuant to the terms of this Contract. If the vendor receives any compensation from another source as a result of conference services performed for the Department, the Contractor will report this compensation to the Contracting Officer and offset its invoice to the Department in an equal amount.

Certification G.26

The offeror has read and understood the above clause in this section.

XYes ☐No

This proposal information **is not** counted toward the 15-page limit of the project narrative.

G.27 EDAR 3452.243-70 Key Personnel (March 2011)

- (a) The personnel designated as key personnel in this contract are considered to be essential to the work being performed hereunder. Prior to diverting any of the specified individuals to other programs, or otherwise substituting any other personnel for specified personnel, the contractor shall notify the contracting officer reasonably in advance and shall submit justification (including proposed substitutions) in sufficient detail to permit evaluation of the

impact on the contract effort. No diversion or substitution shall be made by the contractor without written consent of the contracting officer; provided, that the contracting officer may ratify a diversion or substitution in writing and that ratification shall constitute the consent of the contracting officer required by this clause. The contract shall be modified to reflect the addition or deletion of key personnel.

- (b) The following personnel have been identified as Key Personnel in the performance of this contract:

Labor category	Name
Principal Investigator	Brandon Holding
Contract Administrator – Offeror’s point of contact with signature authority	Brandon Holding

Appendix H

This is not applicable as our business model is already in place.